

Do Local Income Taxes Cause Residential Sorting?

Evidence from Pennsylvania's Piecewise Earned Income Tax

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September 25, 2026

Abstract

Do local income taxes cause residential sorting? Pennsylvania's Earned Income Tax follows a piecewise max-rule where a resident pays the higher of the home-side rates and the workplace rate. The rule creates variation that is used to identify a causal effect. I decompose statutory increases by whether the home-side stack already binds, becomes binding, or never binds, and the never-binding cell is a within-design placebo. I estimate a continuous stacked difference-in-differences design on block-level LEHD/LODES data aggregated to municipalities over 2002–2019. The average causal response to a binding increase is a 0.537 log-point decline per percentage point in the count of workers who both live and work in their own municipality, about a 12% decline at the average cumulative increase of 0.24 percentage points, while the placebo cell is null. The results are robust to multiple tests and specifications. While the headline results originate from a linear model, I run a polynomial and spline specification as well. The two are able to trace the full dose-response, the total and marginal effect at each dose: the effect accumulates steeply across the increases and flattens as they grow. The largest decline comes from the first increase, and each subsequent increase moves fewer workers, consistent with a shrinking pool of residents willing to relocate. All three specifications return similar results. The binding cell passes a set of pre-trend tests and shows no effect at three fake-timing placebo dates. Controlling for municipal, school-district, and county effective property-tax rates shows the property tax's effect is about a third of the EIT's per percentage point.

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1 Introduction

Taxes on wealth and income have been a growing discussion not only at the federal level but across states and local governments in the United States. Much of this debate revolves around growing concerns over inequality and revenue shortfalls. Recently, states such as Massachusetts and California have changed or proposed new taxation rules on the top earners in their states. Massachusetts recently abandoned its flat 5% rate with a 4% tax on income over \$1 million (Vermeer, 2022). Washington State, which has never levied a broad income tax, enacted a 9.9% rate on income above \$1 million in 2026 that does not take effect until 2028 (Washington State Legislature, 2026), and California has proposed a one-time 5% levy on billionaires' net worth for the 2026 ballot (Walczak, 2026). Meanwhile, other states have also followed suit by either raising their income taxes or floating the idea of creating an income tax.

It is not only state governments but local governments as well looking for funding sources as their municipal budgets get squeezed by inflation and shrinking government transfers. Local income taxation is more common than is often recognized: roughly 5,000 jurisdictions across sixteen states levy some form of local income or wage tax (Walczak et al., 2023). New York City has levied its own income tax since 1966 (Walczak et al., 2023), and the current administration has proposed raising it on residents earning more than \$1 million a year to close a projected two-year budget gap (New York City Mayor's Office, 2026). Because the city's taxing authority is delegated by the state, the increase would require approval from the state government, a dependence on state enabling legislation that is characteristic of local income taxation, in general.

Most of this debate happens around the top earners in these states and localities, with much of the common criticism saying that the earners will simply move to avoid the new and higher taxes. That objection is long-standing in public finance, and it is testable: it is a claim about how much of the tax base relocates when a rate changes, and how far a jurisdiction can raise rates before the base it is taxing moves away. The evidence that speaks

to it, however, comes almost entirely from variation in top marginal rates, and that variation does not exist below the state level, where local income taxes are generally flat and fall on every earner alike.

Prediction on the sorting of individuals given variation in taxes has a long history in economic theory. The Tiebout hypothesis predicts that residents sort across jurisdictions which offer different fiscal amenities at different costs and residents will 'vote with their feet' and choose specific bundles of amenities for a given amount of taxation, this is called the Tiebout margin (Tiebout, 1956). This has been tested widely at the state level but less so for municipalities. The literature has focused mostly on state tax adjustments on high-earners (Agrawal & Foremny, 2019; Young & Varner, 2011; Young et al., 2016). The research shows that top-income households are generally mobile and sensitive to higher rates. Some studies have looked at specific higher income groups or niche domains such as star scientists, foreign workers, or even professional athletes (Agrawal & Tester, 2024; Kleven et al., 2020; Moretti & Wilson, 2017). Specifically, Agrawal and Tester (2024) look at professional golfers' response to changes in nonresident income and find that they respond to changes in state taxation of nonresident income by changing where they work rather than where they live.

Few states permit local income taxes and the evidence in the United States is a handful of studies. The closest existing research comes from Yang and Heim (2017), which finds that county-level taxable income in Indiana does not respond to small local rate changes, and Yushkov (2025), who finds county migration flows in Indiana and Maryland that are signed as expected but rarely significant. Pennsylvania's EIT has been studied for compliance rather than for sorting (Bagchi, 2022), and the nearest Pennsylvania precedent is Livas and Jacob (2026) on Philadelphia's wage tax and suburbanization. The strongest sub-state sorting evidence is from Switzerland, where municipalities set their own income tax rates and household location follows the resulting differences (Basten et al., 2017; Schaltegger et al., 2011; Schmidheiny, 2006). The important difference between the local level from the state level is that residence and workplace are often in different jurisdictions, meaning assigning

taxation to residents or non-residents provides important variation for identifying migration effects. Municipalities may have an incentive to tax nonresidents in order to not burden their own residents but are constrained by both needed to raise revenue and not push away those who work but do not live in the municipality. Pennsylvania’s municipalities often have to weigh who to tax, so a household is faced with a set of decisions in order to mitigate their tax bill such changing jobs, schools, or home.

A survey of the literature conducted by Agrawal (2026) highlights the current findings of state and local income taxation and identifies sub-state (county, municipality, school districts) income-tax sorting as one of the field’s open questions. The consensus of the survey is that the literature on local income taxes in the United States is almost non-existent and that what evidence exists is primarily from Switzerland. That matters because Swiss municipalities tax income on a purely residential basis while local systems in the United States assign income using non-residential rules, so the Swiss estimates do easily translate to the US. Whether mobility appears as households moving or as workers moving depends on the source of the taxation itself.

This paper estimates municipal-level migration effects among workers in Pennsylvania from 2002–2019, by merging local income tax data with the LEHD Origin-Destination Employment Statistics (LODES), which links each worker’s residence to their workplace annually and at sub-municipal geography. In Pennsylvania there are roughly 2,500 municipalities and 500 school districts (SD) that set their own Earned Income Tax (EIT) rates (Agrawal, 2026; Bagchi, 2022), producing a deep cross-section of multiple sub-state income-tax variation. The EIT offers two additional dimensions of variation that my empirical design exploits. First, the EIT is set independently by two overlapping rate-setting jurisdictions (municipalities and school districts) and a resident’s home-side tax bill is the sum of the municipal resident rate and the school-district rate. This paper’s primary focus is the municipal resident rate while the school district rate enters the empirical design but is treated as secondary. Second, the EIT operates using a piecewise maximization rule where each worker’s bill is

$T = \max(\tau^{\text{home}}, \tau^{\text{work,muni}})$, where $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$, the sum of the municipal resident rate and the school-district resident rate, and $\tau^{\text{work,muni}}$ is the work rate of the worker’s workplace municipality. The workplace rate is also referred to as the nonresident rate, however, I refer to it as the work rate because the primary outcome variable of is the number of people that live and work within a municipality, which is the only variable that has the cleanest causal identification to the work and home-stack rate changes. A statutory home-side rate change only moves the bill if the home stack is on the binding side of the max. Act 32 of 2008 (effective 2012) consolidated EIT collection into county-based Tax Collection Districts and mandated employer withholding at the higher of the two rates, which made the max-rule mechanically enforced. Bagchi (2022) estimates that Act 32 resulted in a 14% increase in compliance and much higher local revenues. My paper does not analyze compliance but rather estimates residential-sorting and provides additional analysis pre-post Act 32.

Beyond contributing to the general literature on local tax rate changes and residential sorting, this paper makes two methodological contributions. The first uses the piecewise structure of the EIT to split rate-change events by whether the home stack is on the binding side of the maximization function. Increases that move a worker’s bill and increases that leave it untouched are statutory changes of the same size, so the non-binding cell is a placebo the design generates rather than one imposed from outside. This within-design heterogeneity is what Agrawal (2026) identifies as missing from the existing U.S. local-income-tax literature.

The second is the treatment definition. Work on local income taxes codes a rate change as a binary indicator and recovers the average effect of having changed the rate at all (Yang & Heim, 2017; Yushkov, 2025). I use the size of the change, so the estimate is a response per percentage point rather than a response to the average change happened to be in one sample. That matters for two reasons. A binary coefficient is the per-percentage-point effect scaled by the realized distribution of rate changes, so it cannot be compared across settings with different distributions, and pooling changes of different magnitudes under a single indicator treats them as one treatment when they are several, which is the hidden-variations half of

the stable unit treatment value assumption (Imbens & Rubin, 2015, ch. 1). The cost is a stronger identifying assumption, because reading the dose coefficient as a causal response requires trends to be parallel across dose magnitudes and not only between treated and control municipalities. Section 4.5 reports the event study that tests it.

The decomposition is implemented at the municipality-year level in a continuous stacked difference-in-differences design (Cengiz et al., 2019) that uses never-changed municipalities as the control pool. The school-district rate enters the design as a co-treatment on the home-side stack but is not used as a separate unit of analysis. Section 4.2 sets out the binding decomposition and what each group implies for the worker’s bill.

Municipalities that raised their resident rate saw the number of workers who both live and work there fall by about 9%. The binding decomposition shows that this average is carried entirely by the municipalities where the increase actually moved a bill. Where the home stack was already binding (Inc-bind), a typical increase reduces the count of workers who live and work in the municipality by about 10%. Where the increase pushed a home stack that had been sitting at the work rate above it (Inc-becomes), the effect is also about 10%, reached through a larger typical increase and a smaller response per percentage point. Where the rate rose but the home stack stayed below the work rate and no bill moved (Inc-nobind), the estimate is null. That cell is a placebo the design generates rather than one imposed from outside, and its null is the sharpest evidence that the response is to the tax bill and not to rate-setting behavior in general.

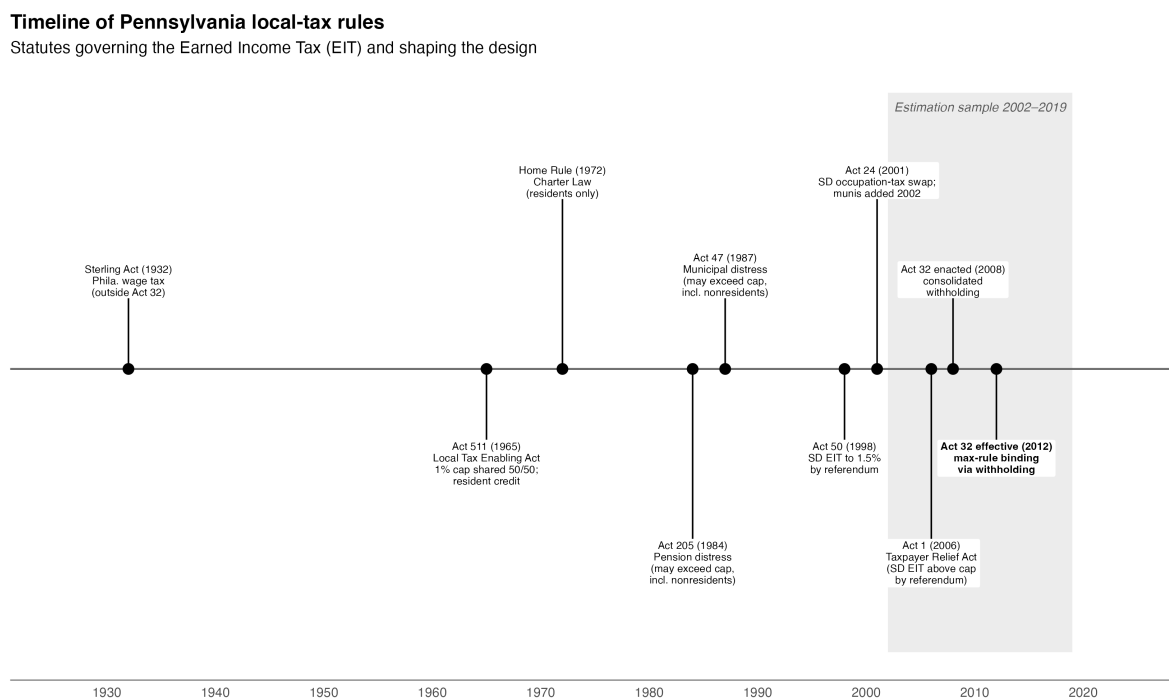
The remainder of the paper is organized as follows. Section 2 describes the institutional setting—the statutes governing the EIT, and Act 32. Section 4.2 introduces the binding decomposition for the EIT. Section 3 describes the data sources: LODES, the DCED rate and millage registers, the STEB common-level ratios, and the ACS income and home-value series. Section 4 explains the continuous stacked difference-in-differences design at the municipality level. Section 5 reports the EIT binding decomposition and the EIT revenue response. Section 6 reports robustness diagnostics.

2 Institutional Background

Pennsylvania has a long history of setting their own income tax rates with approximately 2,500 municipalities and 500 school districts each with the power to levy their own Earned Income Tax on all earned income earned in or by Pennsylvania residents (Agrawal, 2026; Bagchi, 2022). The EIT falls directly on workers and varies across municipal boundaries, creating sharp spatial discontinuities in tax burden that can influence where individuals choose to live and work.

This variation is the product of many decades of government statutes, influencing who may levy the tax, how high the rate may go, and who may be taxed, all of which are key for the design in Section 4. Figure 1 places the many statutes on a timeline that the next section walks through in order.

Figure 1: Timeline of Pennsylvania local-tax statutes; the shaded band is the 2002–2019 estimation window.



2.1 Pennsylvania Earned Income Tax

Local income taxation in Pennsylvania begins with the Sterling Act of 1932 (Sterling Act, 1932), the first local income tax legislation in the country, under which Philadelphia became the first municipality in the United States to impose an income tax in 1939. Philadelphia still operates under the Sterling Act and remains outside the framework governing the rest of the state. Its residents pay the city resident rate regardless of work location, and nonresidents working in the city pay the city work rate. Philadelphia does not enter the estimation sample. Its resident rate declines steadily across 2002–2019, and the design identifies off increase-side events, so the city is dropped along with every other municipality whose rate only falls.

Local income taxation outside Philadelphia was expanded in 1965 under the Local Tax Enabling Act (Act 511; Local Tax Enabling Act, Act 511, 1965), which authorized municipalities and school districts throughout the state to levy an earned income tax on wages, salaries, commissions, and net profits. The act enables a municipality and is not mandatory, therefore a municipality may levy no EIT at all. Where a municipality and an overlapping school district both levy, Act 511 caps their combined resident rate at one percent and splits it fifty-fifty by default, unless the two taxing bodies agree otherwise. School districts may not tax nonresidents. Act 511 also keeps a commuter’s two tax bills from stacking, and it does so by giving the place of residence priority over the place of employment. That priority, rather than any single rate schedule, generates the piecewise structure the paper exploits.

Each Pennsylvania municipality therefore has three EIT rates: the municipality resident rate $\tau^{\text{res,muni}}$ levied on residents of the municipality, the municipality work rate $\tau^{\text{work,muni}}$ levied on workers employed in the municipality, and the school-district rate τ^{SD} levied on residents only. Two governments set these three rates. The municipality sets the resident and work rates, and the school district sets its own. I refer to the two throughout as the rate-setters, so a rate-setter is a government rather than a rate, and the municipality is one rate-setter even though it controls two of the three rates.

That priority means a worker’s total EIT bill is determined by the maximum of the

home-side rate stack and the work-location rate, and the bill is split piecewise between home and work jurisdictions:

$$(\tau^{\text{home}}, \tau^{\text{work}}) = \begin{cases} (\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}}, 0) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \geq \tau_i^{\text{work,muni}} \\ (\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}}, \tau_i^{\text{work,muni}} - \tau_j^{\text{res,muni}} - \tau_j^{\text{SD}}) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} < \tau_i^{\text{work,muni}} \end{cases} \quad (1)$$

where j is the worker's home municipality and i is the work municipality. The home municipality and school district always collect their full stack, and the work municipality collects only the amount by which its nonresident rate exceeds that stack. When the home stack is the larger of the two the work municipality collects nothing (Pennsylvania General Assembly, Local Government Commission, 2025). The worker's total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}}, \tau_i^{\text{work,muni}}). \quad (2)$$

Equation (1) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{work,muni}}$), the worker can lower the bill only by moving their residence to a lower-rate municipality or school district. When the work rate binds, the worker can lower the bill only by changing jobs to a lower-rate work location, while moving homes without changing jobs has no effect. For workers who live and work in the same municipality, the home stack and work rate are both set by that municipality, and the bill is determined by the home stack on the binding side.

A worker who lives in municipality j and moves to a lower-rate municipality j' while keeping the same job at i saves at least the SD rate at j where the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{work,muni}})$, which is at most the old bill τ_j^{home} minus τ_j^{SD} . The empirical design follows this logic directly by using both the treatment $\Delta\tau^{\text{muni}}$ and the binding classification built from the worker's home stack.

State law allows the Act 511 cap to be exceeded in a small number of circumstances. Municipalities adopting a home rule charter under the Home Rule Charter and Optional

Plans Law (Home Rule Charter and Optional Plans Law, 1972) are no longer confined to the statutory limit on taxes levied on their own residents, and some have raised the resident stacked rate above one percent while the nonresident rate may not exceed the statutory limit for nonresidents. Two distress regimes, Act 47 and Act 205, are the exception. A municipality declared financially distressed under the Municipalities Financial Recovery Act (Act 47 of 1987; Municipalities Financial Recovery Act (Act 47), 1987) may petition to raise its EIT above the Act 511 limit as part of an adopted recovery plan, renewable by annual petition, and Act 47 does not bar extending the increase to nonresidents. A municipality certified as having a financially distressed pension system under the Municipal Pension Plan Funding Standard and Recovery Act (Act 205 of 1984; Pennsylvania General Assembly, 1984) holds the same above-cap authority as a pension-recovery remedy, available only once the municipality is already at the maximum rate otherwise allowed, and Act 205 likewise permits extension to nonresidents.

This paper does not estimate the effect of nonresident increases on sorting due to the lack of binding rate changes of the workplace tax over the home-stack. For the work side to bind, a single jurisdiction must exceed the home-stack, which under the standard Act 511 rate structure almost never occurs: across 2002–2019, the work rate is the binding side in roughly 0.2% of municipality-years. The handful of exceptions are the Act 47 and Act 205 distress municipalities just described, the only ones holding statutory authority to exceed the Act 511 cap on nonresidents. Because there are few work-binding events, a work-side binding decomposition is not identifiable.

School districts can go above the cap through referendum. Act 50 of 1998 (Pennsylvania General Assembly, 1998) first authorized a district to levy the EIT at up to 1.5% with voter approval, offset by reducing real estate taxes through a homestead exclusion; those EIT provisions were later repealed, and the Taxpayer Relief Act (Act 1 of 2006; Taxpayer Relief Act (Act 1, Special Session), 2006) is now the route by which a district puts an increase to voters in exchange for homestead property tax relief (Pennsylvania General Assembly, Local

Government Commission, 2025). A third route runs through both levels of government: the Optional Occupation Tax Elimination Act (Act 24 of 2001; Pennsylvania General Assembly, 2001), now Chapter 4 of Act 511, lets a municipality or school district retire its occupation tax and raise the EIT by the revenue that tax had produced, a swap rather than a net increase, and it was extended to municipalities in 2002 (Siger & Shapiro, n.d.). Together with home rule, Act 47 and Act 205 on the municipal side, these are the statutory routes behind the above-cap rates observed in the DCED register.

Act 32 of 2008 (Act 32 of 2008 (Local Tax Enabling Act — Omnibus Amendments), 2008) brought the largest change to the local income tax system since the Local Tax Enabling Act. Signed in 2008 and effective for tax years beginning January 1, 2012, it created 69 county-wide Tax Collection Districts, cutting the number of collectors in the state from roughly 560 and created tax collection committees to oversee collection and distribution within each district. The statute further required uniform regulations, forms and procedures to ensure consistent collections statewide. One district exists per county with two exceptions: Philadelphia is outside the Act 32 framework entirely and has no district, and Allegheny County is divided into four (Pennsylvania General Assembly, Local Government Commission, 2025).

The provision that matters for identification is the withholding rule. Act 32 requires every employer with a Pennsylvania worksite to deduct from each employee the greater of that employee's resident tax or nonresident tax, read off the official tax register that the Department of Community and Economic Development maintains. Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (municipality, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

This had two consequences. First, it made the EIT mechanically binding where the rate changes now appear in every worker's paycheck through employer withholding, regardless of whether the worker resides in the taxing municipality. Second, it eliminated the possibility

of avoidance of the work-location claim through weak enforcement. After Act 32, a worker could no longer effectively avoid the work-rate portion owed to the work municipality.

However, Act 32 did not eliminate all tax savings from residential relocation. The piecewise rule in Equation (1) continues to imply that a worker whose original home stack binds can reduce the bill by moving to a sufficiently lower-stack home jurisdiction where the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{work, muni}})$, which falls below the old home stack τ_j^{home} whenever $\tau_{j'}^{\text{home}} < \tau_j^{\text{home}}$ and the original home stack was the binding side. Act 32 thus changed enforcement and made the max-rule mechanically binding through withholding and it did not eliminate the residential-relocation margin that the mechanism in this paper exploits. Bagchi (2022) highlighted the impact of Act 32 by showing that consolidated employer withholding increased PA EIT compliance thus raising reported earned income in the rate-raising municipality. The present paper complements that work by isolating the residential-sorting response to the same withholding-enforcement mechanism exploiting the variation in whether a given rate change actually moves the worker’s bill (the binding decomposition) and a pre/post-Act-32 split in Section 6 shows the binding-cell sorting response is strengthened (the effect existed prior to Act 32) after Act 32, complementing the compliance-response finding.

Figure 2 documents the institutional break at Act 32. Before 2012, fewer than 60% of PA municipalities reported a work rate to DCED, and after 2012, when consolidated withholding made the max-rule mechanically binding, coverage rises sharply. This is the pre-/post-Act-32 enforcement break used in the heterogeneity test of Section 6.2.

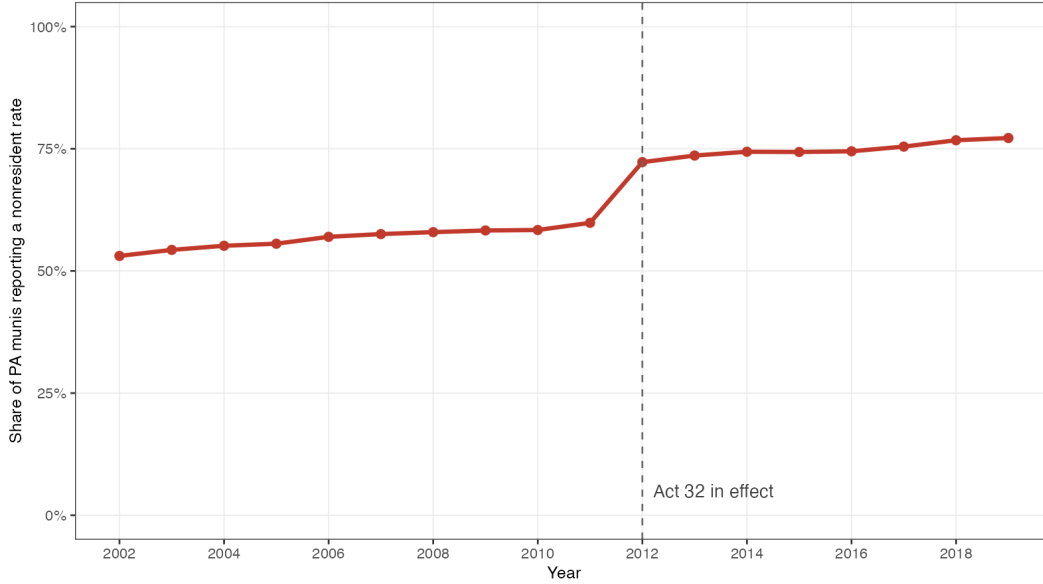


Figure 2: Share of PA municipalities reporting a work rate to DCED, by year. The break at 2012 reflects the Act 32 requirement that employers withhold at the higher of the home resident and work rates.

School districts levy their EIT under the same Act 511 authority. Since every census block sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

3 Data

3.1 Data Sources

The analysis merges the LEHD residence–workplace dataset and the Pennsylvania EIT rate records to create a panel of rate changes and workforce data over space and time. The two datasets do not share common identifiers, and was bridged using TIGER 2020 block geometry, which nests blocks within county subdivision code (cousubs). A municipality–name crosswalk maps each DCED municipality ID to its corresponding COUSUBFP code, matching 2,496 of 2,505 PA municipalities (99.6%). The remaining 9 are small boroughs

that have either dissolved or merged during 2002–2019 and are dropped from the analysis.

The raw 2002–2019 municipality panel holds 45,291 muni-years across 2,559 municipalities. I first drop municipalities whose rate history moves in both directions over the sample, because a municipality that raises and later cuts its rate cannot be assigned a single binding-cell classification, along with municipalities whose first event is a decrease, which are outside the paper’s scope and serve as neither treated units nor controls. The test is applied to each of the three rates a municipality faces, its own resident rate, its school district’s resident rate, and its own nonresident rate, so a municipality with a clean resident-rate history is still removed if one of the other two reverses. The nonresident rate accounts for 554 of the 795 municipalities dropped at this step, and because that rate is not itself estimated here the restriction is conservative rather than identifying. That leaves 31,744 muni-years across 1,764 municipalities. I then drop muni-years missing any fiscal control, which is the sample the regressions select through listwise deletion, leaving 24,229 muni-years across 1,763 municipalities. The second restriction removes years rather than places, since AFR filing coverage is uneven early in the panel. Because the outcomes are logged, each regression drops the additional muni-years in which the relevant count is zero, so per-outcome sample sizes appear in Table 1 row by row.

The residence–workplace data come from LEHD LODES Version 8, 2002–2019 (U.S. Census Bureau, 2024b). The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality and is calibrated so that aggregation to tract and larger geographies approximates ground truth (Graham et al., 2014). The sample period of 2002–2019 reflects LODES coverage beginning in 2002 and 2020–2023 to not include disruptions in the data resulting from the COVID Pandemic.

The origin–destination file records job counts by residence block and workplace block, which I aggregate to the municipality and then transform into the four outcomes used. The first variable, Live-and-work, are workers whose residence and workplace municipalities are the same, I argue this is the outcome a resident rate change acts on most directly. The

second outcome variable is in-commuters which are jobs located in the municipality but held by residents of another. Third, total jobs is the total number of jobs in a municipality, a proxy for the measure of how well a municipality’s economy is functioning. Lastly, live-and-work share divides live-and-work workers by total jobs, so its denominator is the local job base rather than the resident labor force. A coefficient on that share therefore answers how the composition of a municipality’s workforce shifts, and it can move because residents leave or because in-commuting grows.

Municipal and school-district earned income tax (EIT) rates come from the Pennsylvania Department of Community and Economic Development (DCED) MunStats portal (Pennsylvania Department of Community and Economic Development, 2024a), and is scraped for annual values. In some DCED reports, the flat-dollar Local Services Tax is filled in the same field as the EIT in some years, so rates above 5% are treated as data-entry errors and dropped except for Home Rule cities. The municipality and SD assignment uses the SD rates reported for each muni in the DCED rate panel (97.6% of municipalities are single-SD), so each muni-year carries a single τ^{SD} co-treatment. For 2.4% of municipalities that have multiple school districts that overlap within its borders I take the max of the set.

Municipal fiscal controls come from the PA DCED Statewide Municipal Annual Financial Report (AFR) database (Pennsylvania Department of Community and Economic Development, 2024b). School-district fiscal controls come from the Census Annual Survey of School System Finances (F-33) (U.S. Census Bureau, 2024a), and the NCES LEAID-to-DCED SD ID crosswalk covers all 500 school districts. Municipality regressions use four DCED AFR shares, three of expenditure (police services, streets, debt service) and one of revenue (inter-governmental revenue), together with the three effective property-tax rates described below. All fiscal controls are share-based and time-varying. Two AFR shares are deliberately left out. First, the EIT-revenue share is a function of the EIT rate, so it sits between the treatment and the outcome and conditioning on it absorbs part of the effect being estimated. Second, the real-estate-tax share is a revenue-side proxy for the same property burden that

the effective property rates measure directly so it is also not used in the estimation.

The property tax for municipal, school-district, and county real-estate millage come from the same DCED MunStats register, and is scraped for annual values (Pennsylvania Department of Community and Economic Development, 2024a). The county tax register reports general-purpose and total millage for all 66 assessing counties.

Millage is the statutory rate, one mill being one dollar of tax per thousand dollars of assessed value. Assessed value is not market value. Pennsylvania counties assess at widely different fractions of market value and reassess rarely, so identical millage implies very different burdens across counties and the statutory rate is not comparable on its own. The common level ratio bridges the two. Pennsylvania law defines it as the ratio of assessed value to current market value prevailing in a county, determined annually by the State Tax Equalization Board from validated sales data (Pennsylvania Department of Community and Economic Development, 2025). County ratios come from the Department of Revenue’s historical factor series (Pennsylvania Department of Revenue, 2025). The property tax is therefore calculated as:

$$\tau^P = \frac{\text{mills}}{1000} \times \text{CLR}, \quad (3)$$

I compute each jurisdiction separately since the municipality is the estimating unit and the three governments set rates independently.

However, the tax calculation is not always accurate due to infrequent assessments. In a reassessment the county revalues every property at current market value, so assessed value rises to the market value and the ratio between them should rise with it. Because the tax base has grown, each government lowers its millage in proportion, and the effective tax bill the property owner actually pays should not change. Where a government’s millage was never restated on the new basis, the old millage is paired with the new ratio, and the computed burden jumps by the size of the reassessment and stays high for the rest of the panel.

I correct the millage rather than dropping the observations. Two rules do the work. The first looks for the reassessment. A reassessment is recognized by the ratio moving to one,

which is what assessing at full market value means, rather than by how far the ratio moved, because ordinary market drift moves it every year and some genuine reassessments move it very little. In a reassessment year a government that failed to restate its millage is one whose millage did not fall when the base grew, and the size of that failure is also the correction. The second rule looks across municipal governments. Every municipality in a county shares one ratio, so a municipality whose computed burden sits far above its neighbors' for most of the panel is on the wrong basis even when no reassessment inside the sample reveals it. The correction touches a small share of observations, leaves no municipal or county burden at an implausible level, and makes the series smoother from year to year. Every corrected value is written to an audit file recording what it was, what it became, and which rule fired.

The school-district layer resists both rules and some districts remain implausibly high. Those are transcription errors in the dataset, and districts whose millage never matched their county's basis anywhere in the sample meaning there is no break for the rules to find. I leave them rather than dropping the variation. The school district enters the municipality-level specification only as a control, and the headline estimate barely moves when it is left out altogether, so what remains does not affect the results.

3.2 Descriptive Statistics

This section presents the summary statistics on the estimation sample. The table reports the tax rates, the four outcomes and the fiscal controls in three panels.

Table 1: Estimation-sample summary statistics, muni-year (2002–2019)

Variable	N	Mean	SD	Min	Max
<i>Tax rates (percentage points)</i>					
Resident EIT rate τ^M	24,075	0.52	0.13	0.00	2.50
School-district EIT rate τ^{SD}	24,075	0.64	0.30	0.00	2.05
Home stack $\tau^M + \tau^{SD}$	24,075	1.16	0.31	0.00	3.40
Nonresident EIT rate $\tau^{nr,M}$	23,915	0.89	0.22	0.00	1.95
Effective property rate, muni	24,229	0.20	0.20	0.00	3.80
Effective property rate, school district	24,229	1.59	0.87	0.00	8.34
Effective property rate, county	24,229	0.44	0.23	0.00	3.74
<i>Outcomes</i>					
Live-and-work workers	23,890	207.29	734.74	0.00	21863.00
Share live-and-work in muni (%)	23,890	9.40	7.65	0.00	100.00
In-commuters	23,890	1681.75	3801.53	0.00	55208.00
Total jobs	23,890	1889.04	4364.96	1.00	62714.00
<i>Fiscal controls (percent of total)</i>					
Police, share of expenditure	24,229	9.29	10.97	0.00	64.03
Streets, share of expenditure	24,229	32.86	20.89	0.00	97.71
Debt service, share of expenditure	24,229	4.89	7.70	0.00	81.04
Intergovernmental, share of revenue	24,229	23.31	16.97	0.00	100.00

Notes: Estimation sample: 24,229 muni-years across 1,763 municipalities, from a raw panel of 45,291.

Each row reports its own non-missing N . Rates in percentage points. Effective property rates are millage times the county common-level ratio, after the repair in Section 3.1.

Table 1 shows that the resident rate is tightly clustered near the Act 511 default at $\tau^M = 0.52\%$ with a standard deviation of 0.13. The maximum is 2.50%, which is well below Philadelphia (4.33%) which is dropped from the dataset as stated above. The SD

rate has more than twice the muni-rate dispersion (0.30 against 0.13) and with an average (0.64%) above the municipality (0.52%), the school district is a larger statistical lever of the two rate-setters, the municipality and the school district. The home stack averages 1.16%, above the 1% Act 511 cap, due to Act 1 SD increases and Home-Rule muni rates pushing the combined level past the ceiling. The nonresident rate averages 0.89%, well above the home stack's municipal component and close to the stack as a whole, which is what makes the max-rule bind for so many workers. It is also the flattest of the four, with a standard deviation of 0.22 and a maximum of 1.95%, because municipalities have far less latitude over the rate they charge commuters than over the one they charge their own residents.

The property taxes are much larger than the income tax. The school district alone levies 1.59 percentage points of market value on average, three times the resident EIT rate, and the three levels together come to roughly 2.2 points against a home stack of 1.16. Their standard deviation is also wider, and even the smallest level varies more across municipalities (0.23 for the county) than the resident EIT rate does (0.13). A municipality considering an EIT change is therefore weighing the adjustment of other local taxes on that are levied on same household, which is why the property rates enter as controls.

All four outcomes are built from the same LODES origin-destination primitive, a count of jobs located in a municipality classified by where the worker lives. Total jobs is every job located in the municipality. Work in own municipality (jobs in the municipality whose holder also lives there) is the count of workers who both live and work in it. In-commuters (jobs in the municipality whose holder lives elsewhere) is the complement, so total jobs is exactly the sum of the two. The share live in municipality (work in own municipality divided by total jobs) is the fraction of local jobs held by residents.

The two outcomes the paper leans on are the count of workers who both live and work in the municipality and that count as a share of local jobs. The average municipality has 207 such workers, but the distribution is extremely skewed, with a standard deviation of 735 and a maximum near 22,000, so the logged count is the estimating outcome. The share averages

9.4%, meaning fewer than one in ten jobs in a typical municipality is held by someone who lives there. Residential sorting therefore operates on a thin margin. The remaining two outcomes are proxies for the labor market of these municipalities rather than sorting. In-commuters average 1,682 per municipality and total jobs 1,889. Both are reported because a change in the share is now interpretable alongside the flows that generate it. They are also valuable to include because they may indicate an economy in decline even if there is a statistically significant effect on the primary two outcome variables.

Two consequences of this construction matter for reading the results. The four outcomes are not four independent measurements. In-commuters make up roughly nine tenths of total jobs, so the total-jobs coefficient is close to a restatement of the in-commuter one, and the share carries total jobs in its denominator. A response that moved residents and in-commuters by the same proportion would leave the share unchanged, so the share only moves when the two counts move differently. Separately, the logged outcomes are undefined where a count is zero, so the log regressions run on slightly smaller samples than the share regression, which keeps those municipality-years.

The fiscal controls describe how a municipality spends and funds itself. Streets are the largest single expenditure category at 32.9% on average, police the most variable at 9.3% with a standard deviation of 11.0, and debt service the smallest at 4.9%. On the revenue side, intergovernmental transfers average 23.3%, so roughly a quarter of the typical municipal budget arrives from another level of government. These shares are time-varying within municipality and are included to absorb fiscal pressures that might move both the rate and the population.

Table 2: Cell counts: muni resident-rate side, 2002–2019

Cell	Munis	Muni-years	Mean dose (pp)	Mean dose at +5 (pp)	Extreme dose (pp)
Never-changers	1,686	23,165	0.00	0.00	0.00
Inc-bind	24	334	0.21	0.24	0.85
Inc-becomes	39	545	0.27	0.38	1.25
Inc-nobind	14	185	0.34	0.43	0.65
Pooled Increase	77	1,064	0.27	0.34	1.25

Notes: Estimation sample, as in Table 1. Municipalities are classified on their first resident-rate change.

Mean dose averages the cumulative rate change over every muni-year in the cell, including pre-event years where it is zero. Mean dose at +5 is the eventual cumulative increase, five years after the event, and is the figure used to convert coefficients into effects in Section 5.

Table 2 reports the cell assignments on the muni resident-rate side. Within the estimation sample, 1,686 never-changers form the NC control pool and only 77 municipalities enter one of the three increase cells, split as 24 Inc-bind, 39 Inc-becomes, and 14 Inc-nobind. Inc-becomes shows a larger average dose than Inc-bind, indicating that crossing the binding threshold typically requires a bigger statutory move than slowly inching up an already-binding rate.

The doses shrink as the cell becomes more binding. Inc-nobind, the cell in which the rate rose but no bill moved, carries the largest average dose of the three at 0.34, against 0.27 for Inc-becomes and 0.21 for Inc-bind.

Figure 3 reports the timing of municipal resident-rate changes across the panel. Increases are concentrated in the first half of the sample and taper off after 2009. Of the 77 treated municipalities in the estimation sample, 60 have event years before 2012. Most of the identifying variation therefore predates Act 32, which both indicates that the effects recovered are driven by smaller increases and warrants the pre-post 2012 decomposition reported in Section 6.

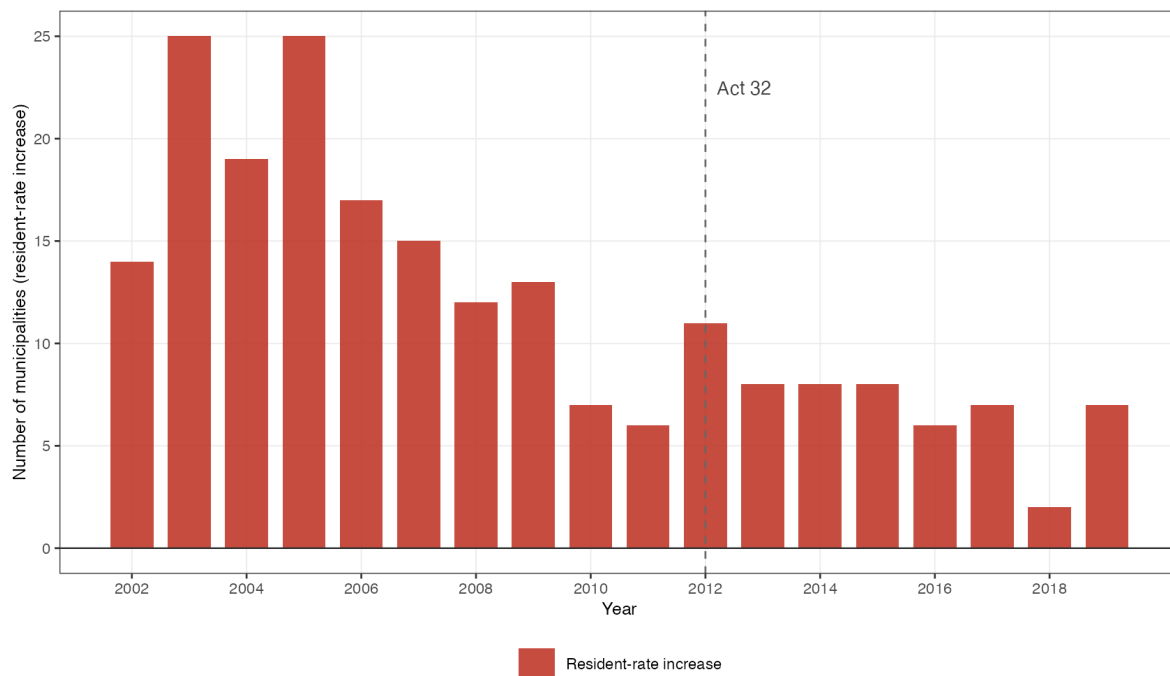


Figure 3: Number of PA municipalities changing their resident rate, by year. Increases shown above zero; decreases shown below. Dashed line: 2012, when Act 32 consolidated withholding.

Lastly, figure 4 shows the spatial distribution of the rate changers in my sample. Most rate changers occur along the east coast, but some are also scattered within the interior of the state. This spatial distribution highlights the value of future work to understand the effects when two neighboring municipalities raise or lower rates across time.

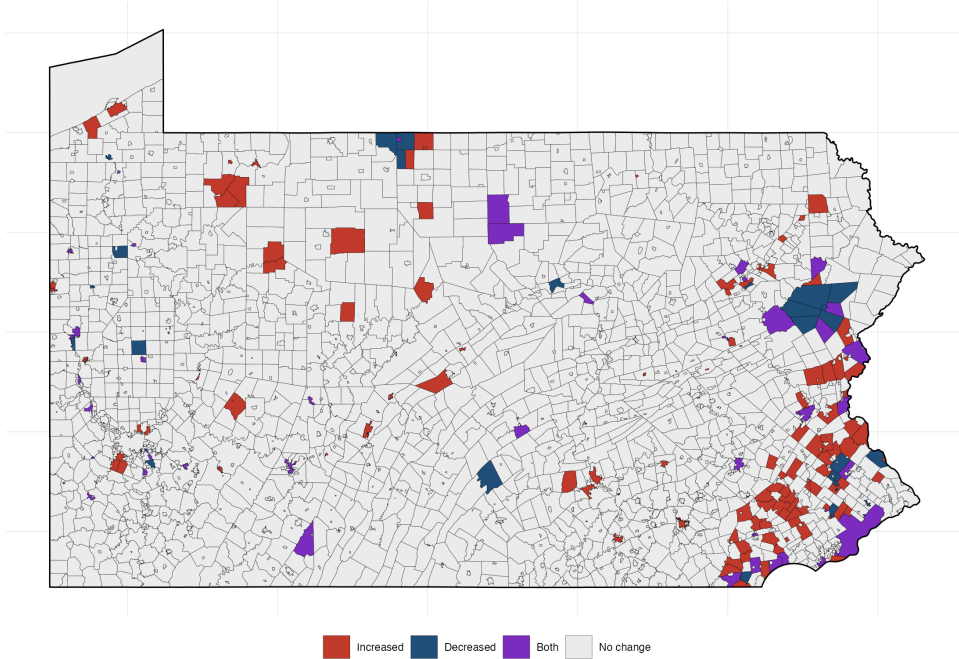


Figure 4: Pennsylvania EIT rate changes, 2000–2019. Red = increased; blue = decreased; purple = both; grey = no change.

4 Empirical Strategy

I estimate two specifications using a continuous stacked difference-in-differences design (Cengiz et al., 2019). First, I run a pooled-increase specification (Section 4.3) that groups all rate-increase events into a single group. Second, I use the piecewise function in defined in Section 4.2, isolating the rate changes that move the worker’s bill from those that do not, called the binding decomposition (Section 4.4). Both specifications are estimated at the municipality-year level. I also estimate an event study for both specifications. Pre-trends are tested separately by a realized-dose Sun–Abraham event study (Section 4.5). This section focuses on the causal parameters estimated, the stacked difference in differences designs, identification, and diagnostics.

4.1 Causal Parameters: ACR, ACRT, and ATT

This paper estimates a stacked continuous difference-in-differences design, specified in detail below. This section sets out what a continuous treatment buys and what it costs. Continuous DiD is a fast moving corner of the econometrics literature, and recent work has sharpened considerably what has to hold before a given causal parameter can be recovered.

In a standard 2 x 2 design the treatment is binary and, under the usual assumptions, the coefficient is the average treatment effect on the treated. A continuous treatment changes that, because the effect now depends on how large the rate change was. One coefficient becomes three related objects. Where the binary case asks whether a municipality raised its rate, the continuous case asks by how much, and the additional variation supports a greater number of parameters to estimate.

The clearest way to understand the three is as a curve. In Baker et al. (2025), the authors plot the effect against the dose d , running from zero upward. This paper hypothesizes a negative effect, so the same curve is drawn declining rather than rising, the mirror image of theirs. Each point on the curve is an $ATT(d)$, the total effect of a rate change of size d measured against never-changers, so a municipality that raised its rate by d percentage points ends up $ATT(d)$ log points away from one that did not. The slope of the curve is $ACRT(d)$, the average causal response on the treated, which asks what one further percentage point does to a municipality that has already raised its rate by d . The average of that slope across the rate changes municipalities actually chose, weighted by how common each one is, is the average causal response (ACR). A linear specification returns exactly this last quantity (Baker et al., 2025; de Chaisemartin et al., 2024). It is the headline estimand of the paper, and it plays the role for a continuous treatment that the ATT plays for a binary one.

The dose-response figures in this paper are fit to a natural spline in the dose, with interior knots at the terciles of the positive treated doses and boundary knots at zero and the largest observed dose. The spline is a visual device rather than an estimating equation, and it is deliberately more flexible than the data can support so that any bending is visible if it is

there. Some of it is, but not where it would matter most. A quadratic in the dose rejects a constant marginal effect for the pooled increase group on the count of workers who live and work in the municipality ($p = 0.004$) and on their share of local jobs ($p = 0.024$). Fitted within cells the rejection belongs to the partial pass-through group ($p = 0.012$) and not to the binding group, where the squared term is nowhere near significant on any outcome (p between 0.60 and 0.78). Pooling groups whose linear slopes differ will produce apparent curvature even when each group is straight, so the pooled rejection is at least as much a statement about mixing binding states as about the shape of any one response.

It is worth being explicit about what this means for the estimated model. Under a linear specification the marginal effect cannot vary, so $ACRT(d)$ is constant and equal to the ACR at every dose. The two parameters are the same number, and the distinction between them is visible only once the dose is allowed to enter flexibly. That is the work the spline figures do, and it is the reason they are reported alongside a linear headline rather than instead of one.

Curvature does not undermine the headline estimand, because the ACR is defined as a dose-weighted average of marginal effects that vary with dose. It is the case the ACR exists to summarise, and a linear specification recovers that average whether or not the underlying curve bends. What curvature rules out is reading the coefficient as a constant marginal effect. The reported number is the average response across the rate changes municipalities actually chose, not the effect of one further percentage point at any particular starting point.

Section 6.1 therefore re-estimates the effect with the de Chaisemartin and D’Haultfœuille estimator, which recovers a weighted average of marginal effects without imposing linearity and without the negative weights a two-way fixed effects regression can place on some municipality-year cells when effects are heterogeneous. Agreement between the two is evidence that the linearity restriction is not driving the result.

Besides the general assumptions of a DiD model, three assumptions are relied on for causal interpretation. The first is the strong parallel trends assumption which states that parallel

trends must hold not only between municipalities that raised their rate and those that did not, but across the size of the increase, so that municipalities choosing large increases would have followed the same path as those choosing small ones had neither acted. The strong parallel trends assumption is what permits interpreting the slope of the dose-response curve as causal (Callaway et al., 2024). Section 4.5 tests it directly by using an event study to tag each municipality with the rate change it eventually takes and asks whether municipalities headed for large changes were already on different trajectories. The second assumption is no anticipation, meaning that outcomes in the years before a rate change do not already respond to it. Rate changes are set in public budget processes with some lead time, so this is an assumption about how far in advance workers act rather than about whether the change could have been foreseen at all. Section 4.5 tests it jointly with parallel trends, since pre-event coefficients respond to either failing. Third, the stable unit treatment value assumption (SUTVA), which has two components. The first is the standard assumption of no spillovers from the treated unit to the control group or vice-versa. The second is that there cannot be hidden variations of treatment, such that a given treatment level does not stand for several different interventions. It is the second component that a continuous treatment satisfies and a binary coding of the same policy does not. In the first requirement a household leaving one municipality may move to a neighboring one that may sit in the control pool, which would bias the effect toward zero. In the second component, the dosage is no more or less than what is observed such that if a municipality raised its EIT by half a percentage point it did not in fact raise it by one, nor do we assign a binary treatment where every increase is condensed into a single (binary) treatment where very different magnitudes would be treated as the same intervention (Imbens & Rubin, 2015). Lastly, stacking adds one further requirement, that never-changing municipalities are clean controls throughout each event window, which the classification enforces by admitting only municipalities whose rates never move over the whole sample.

4.2 The Binding Decomposition

The piecewise tax allocation in Equation (1) drives the causal identification of my empirical design. A worker’s bill is the larger of two rates. The home rate $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$ is set by where the worker lives such as the municipality and the school district together. The work rate $\tau^{\text{work,muni}}$ is set by the municipality where the worker is employed and applies to people who work there without living there. As Equation (1) sets out, the home side always collects its full stack and the work side collects only the amount by which its rate exceeds that stack, so the worker’s total is the larger of the two. A rate change therefore moves that total only if it moves the side that is already larger, or if it lifts the home rate past the work rate.

An increase can therefore land in one of three ways, and these are the groups the design estimates. Where the home rate already exceeds the work rate, the increase passes through in full and the bill rises by the whole rate change (Inc-bind). Where the home rate starts below the work rate and ends above it, only the portion above the work rate reaches the worker, so the bill rises by less than the rate change (Inc-becomes). Where the home rate rises but stays below the work rate throughout, the bill does not move at all (Inc-nobind). Municipalities that never change a rate form the control pool (NC). Three mirror decrease groups exist by construction but are not estimated here, because the decrease cohort is thin (4 Dec-bind, 3 Dec-becomes-non, 11 Dec-nobind) and the realized-dose pre-trend rejects parallel trends on every decrease outcome.

This design produces three testable predictions across coefficients. Each coefficient should be descending in effect size from Inc-bind where the effect is the largest and exceeds Inc-becomes effects, which in turn exceed Inc-nobind effects, with the last close to zero and serving as the within-design placebo. A tax salience argument would predict a larger Inc-becomes response than pass-through alone implies, since that is the group where the home stack crosses the work rate and the work municipality stops collecting from these workers altogether, which is a discrete change in who taxes them even though the total moves only

partially. However, the Inc-bind is taxed the most already and adding an additional tax on top of their already high tax may prove to be a stronger effect than Inc-becomes. If Inc-nobind, being the in-design placebo, has a larger and statistically significant effect, the response is not the binding rule, because an Inc-nobind increase leaves the worker’s total bill untouched. Some other aspect such as municipal spending on specific goods and services may be driving the outcome. In the decomposition model the piecewise function predicts that only a subset of increases should generate a response, however, I also run a pooled model where every increase is considered treatment event to understand if the simple case of increases induce migration.

4.3 Pooled Specification

In this section I report an aggregate specification that pools the treated Increase cells by rate-setter into a single group Inc (any resident rate increase), with never-changers (NC) as the control group. The pooled regression answers, what is the average per-percentage-point (pp) effect of a rate hike on residents, ignoring whether or not the tax increase binds?

For each unit-time observation (i, t) on the stacked panel, the per-pp pooled regression is

$$\begin{aligned}
Y_{\tilde{it}} = & \alpha_{\tilde{i}} + \gamma_{\tilde{t}} + \beta_{\text{Inc}}^M \tau_{it}^M \mathbf{1}\{d_i^M = \text{Inc}\} \\
& + \beta_{\text{Inc}}^{SD} \tau_{it}^{SD} \mathbf{1}\{d_i^{SD} = \text{Inc}\} \\
& + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{\tilde{it}}.
\end{aligned} \tag{4}$$

The dependent variable $Y_{\tilde{it}}$ is the outcome of interest. At the municipality level the outcomes are the logs of live-and-work workers (residents employed in their own municipality), in-commuters, log total jobs, and the share of workers living in their own municipality. The unit i is a municipality and t a calendar year. The subscripts $\tilde{i} = (i, c)$ and $\tilde{t} = (t, c)$ are the unit-by-stack and year-by-stack groupings where each treated unit i appears in exactly

one sub-experiment (its first-event cohort c), while each NC unit appears in every cohort. Cohort c is defined as the first year that the rate-setter changed τ . Each sub-experiment runs on the window $t \in [c - 5, c + 5]$, so a cohort contributes eleven years centred on its own event. The fixed effects $\alpha_{\tilde{i}}$ and $\gamma_{\tilde{t}}$ are for each unit and cohort pair. The directional grouping $d_i^s \in \{\text{NC}, \text{Inc}\}$ classifies unit i 's rate trajectory on rate-setter $s \in \{M, SD\}$, with NC the omitted reference as they are the control group. The treatment variable τ_{it}^s is the statutory rate itself, in percentage points, entered in levels. The unit fixed effects $\alpha_{\tilde{i}}$ absorb each unit's own rate level, so the coefficient is identified from within-unit movement in the rate and reads as a response per percentage point of rate change. The coefficient of interest is β_{Inc}^M , which is interpreted as the per-percentage-point effect on Y of a municipal resident-rate increase. Because the rate enters linearly, it is an average causal response, weighted by how common each rate change is. β_{Inc}^{SD} is its school-district counterpart, included so the municipal effect is estimated separately of what the school district did instead of absorbing it. This design is close to Yushkov (2025), who studies local income tax changes with a binary treatment indicator. I extend that approach to the continuous variation, using the size of each rate change rather than its occurrence. The control vector $\mathbf{X}_{it}$ holds four municipal fiscal shares (police services, streets and debt service as shares of expenditure, intergovernmental revenue as a share of revenue) and the three effective property-tax rates described in Section 3.1. ϕ is the coefficient vector on these controls and $\varepsilon_{\tilde{it}}$ is the error term. Standard errors are clustered at the municipality, the jurisdiction that sets the rate and therefore generates the variation being used.

4.4 Binding Decomposition

The binding decomposition is the main specification of the paper, and it is Equation 4 with now decomposed by the binding principles outlined above. The one Inc group is split into the three binding groups of Section 4.2, so Inc-bind, Inc-becomes and Inc-nobind each receive their own coefficient and NC remains the control. Decrease groups are not estimated, for

the reasons given in Section 4. Everything else is unchanged, including the panel, the fixed effects, the rate in levels, the controls, and the clustering.

$$\begin{aligned}
Y_{it} = & \alpha_i + \gamma_t + \sum_{k \in \mathcal{C}_{\text{NC}}} \beta_k^{\text{muni}} \tau_{it}^{\text{muni}} \cdot \mathbf{1}\{\text{cell}_i^{\text{muni}} = k\} \\
& + \sum_{k \in \mathcal{C}_{\text{NC}}} \beta_k^{\text{SD}} \tau_{it}^{\text{SD}} \cdot \mathbf{1}\{\text{cell}_i^{\text{SD}} = k\} \\
& + \mathbf{X}_{it}' \boldsymbol{\phi} + \varepsilon_{it}.
\end{aligned} \tag{5}$$

For each rate-setter s and outcome Y the index k runs over the three estimated groups, $\mathcal{C}_{\text{NC}} = \{\text{Inc-bind}, \text{Inc-becomes}, \text{Inc-nobind}\}$, and cell_i^s is the group municipality i was assigned on rate-setter s by its first rate change. That label is fixed once assigned and does not vary over time.

The indicator $\mathbf{1}\{\text{cell}_i^{\text{muni}} = k\}$ equals one when municipality i belongs to group k and zero otherwise. The municipality's own rate is then multiplied by the coefficient belonging to its group k . For example, an Inc-bind municipality contributes $\beta_{\text{Inc-bind}}^{\text{muni}} \tau_{it}^{\text{muni}}$, an Inc-becomes municipality contributes $\beta_{\text{Inc-becomes}}^{\text{muni}} \tau_{it}^{\text{muni}}$, an Inc-nobind municipality contributes $\beta_{\text{Inc-nobind}}^{\text{muni}} \tau_{it}^{\text{muni}}$, and a never-changer is set to zero because it is the comparison group. The sum is therefore a compact way of writing three separate slopes, one per group, estimated in the same regression against a shared set of controls. The second summation does the same on the school-district side.

I interpret each β_k^s as an average causal response, the dose-weighted average per-percentage-point effect of rate s on Y within group k , rather than the marginal effect at any single dose (Baker et al., 2025; de Chaisemartin et al., 2024). Section 6.1 shows this ACR is recovered with non-negative weights and is not distorted by dose curvature. $\beta_{\text{Inc-nobind}}^{\text{muni}}$ is the within-design placebo, where the increase leaves the bill untouched, so it should be statistically indistinguishable from zero.

4.5 Realized-Dose Event-Study Diagnostic

Equations 4 and 5 cannot test for pre-treatment leads directly because τ_{it}^s does not move before the event, so once the unit fixed effect removes each unit's own level there is no pre-event variation left for a lead coefficient to be estimated from. I therefore run a separate realized-dose Sun–Abraham event study on the same stacked panel to test for strong parallel trends. I design the event study as recommended for continuous-treatment DiD designs by Callaway and Sant’Anna (2021) and de Chaisemartin et al. (2024) which states that units with larger eventual rate changes should not be on differentially sloped pre-event trajectories relative to NC controls. For each cell on rate-setter s I restrict the panel to units in (NC, cell) and run:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \beta_k^M \mathbf{1}\{\ell_{it} = k\} \tilde{D}_i + \mathbf{X}_{it}' \boldsymbol{\phi} + \varepsilon_{it}. \quad (6)$$

Two variables are new relative to Equation 5. The event-time index $\ell_{it} = t - c_i^M$ counts the years between calendar year t and unit i 's first muni-rate event year c_i^M , with $\ell = -1$ (one year before the event) as the omitted reference year and in the sum below k denotes a value of ℓ , an event-time, in contrast to Equation 5, where k indexed the binding cells. The realized-dose tag $\tilde{D}_i = \tau_{i,c+5}^M - \tau_{i,c-1}^M$ is a single number per unit where i 's eventual cumulative rate change is the observed rate five years after its first event minus the rate the year before and held fixed across all event-time years rather than moving only at the event as τ_{it}^M does in Equation 5. It enters the regression only through the interaction $\mathbf{1}\{\ell_{it} = k\} \tilde{D}_i$, which equals $\tilde{D}_i$ in the single year unit i is at event-time k and zero in all its other years because that varies within unit, the unit fixed effect does not absorb it. The outcome Y_{it} , the unit-by-stack and year-by-stack fixed effects, the control vector $\mathbf{X}_{it}$, the units of observation, and the policy-jurisdiction clustering are the same as in Equation 5. I report the joint Wald p -value as a single pre-trend statistic for each cell-outcome combination.

5 Results

I report the results in three components. First, Section 5.1 presents the pooled Inc model, which estimates the effect of a rate-increase on worker migration without distinguishing whether the tax change binds. Second, Section 5.2 reports the results of the increase through binding decomposition model at the municipality level. Third, Section 5.3 present the results of the event study for the pooled and binding decomposition models. Dec cells are not estimated because they are too thin of a sample.

5.1 Pooled Results

I first report the pooled-Increase specification. The pooled coefficients estimate the average per-pp dose-response of a municipality-rate change and an SD-rate change.

Table 3 reports the per-pp pooled coefficients. The table reports the set of tax rate variables such as the municipality resident-rate Inc effect β_{Inc}^M and the school-district-rate Inc effect β_{Inc}^{SD} . It also reports the fiscal controls such as shares of expenses and property tax rates.

Table 3: Pooled Increase headline — municipality level (per pp)

Outcome	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Municipality rate (τ^M)				
$\Delta\tau > 0$ (Inc)	-0.021* (0.011)	-0.284*** (0.052)	-0.220*** (0.052)	-0.248*** (0.052)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.007 (0.006)	0.114 (0.074)	0.113 (0.070)	0.122* (0.069)
<i>Fiscal controls</i>				
police-services share	0.002 (0.007)	0.025 (0.092)	-0.090 (0.096)	-0.089 (0.093)
streets share	0.001 (0.003)	0.051 (0.041)	0.018 (0.041)	0.021 (0.041)
debt-service share	0.004 (0.005)	-0.046 (0.054)	-0.026 (0.061)	-0.018 (0.059)
intergovernmental-revenue share	0.008** (0.004)	0.040 (0.045)	0.025 (0.050)	0.034 (0.049)
muni property eff. rate (pp)	-0.013** (0.005)	-0.213*** (0.051)	-0.110** (0.045)	-0.129*** (0.044)
SD property eff. rate (pp)	0.001 (0.001)	0.017** (0.009)	0.009 (0.009)	0.010 (0.009)
county property eff. rate (pp)	0.001 (0.002)	0.031 (0.024)	0.011 (0.022)	0.013 (0.022)
N	187,815	171,451	187,807	187,815
R^2	0.826	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001
Joint Wald p (leads)	0.259	0.029	0.060	0.103

Notes: Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled Inc treatments jointly; the reported coefficients are β_{Inc}^M and β_{Inc}^{SD} . * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A one point increase reduces the count of workers who both live and work in the municipality by 0.284 log points, a decline of 24.8 percent ($e^{-0.284} - 1$). In-commuters fall by 19.7 percent and total jobs by 22.0 percent, each statistically significant to the one percent level. The live-and-work share is not a logged outcome but instead a percentage. The coefficient of the live-and-work share is -0.021 and is interpreted as a fall of 2.11 percentage points, significant only at the ten percent level.

In my sample a full percentage point change is larger than the average change. The largest cumulative increase in the sample is 1.25 points, only four municipalities reach a full point or more, and the average five years after the first event is 0.34 (Table 2). Evaluating the results at that average increase, the resident workforce falls by 9.2 percent ($e^{\hat{\beta} \times \bar{d}} - 1 = e^{-0.284 \times 0.34} - 1 = -0.092$), in-commuters by 7.2 percent, total jobs by 8.1 percent, and the share by 0.72 points. I report both interpretations because the first is the direct interpretation of the coefficients and the second is a weighted estimate by the tax change municipalities actually made.

School-district rate increases carry small positive coefficients on every outcome. Only the logged total jobs is statistically significant to the ten percent level. The standard errors indicate I cannot be confident in the estimated coefficient, however the estimates' direction (positive) points to two possible reasons: first, schools are seen as positive institutions and it is well understood that families and individuals want to sort into good school districts. Raising the school district tax may create a positive loop whereby it signals that further investment into an already well sought after amenity further incentivizes migration into those districts.

The property-tax estimates vary in sign depending on the level of government levying it. They are control variables and are not to be interpreted as causal estimates. A one percentage point higher municipal effective property rate is associated with 0.213 fewer log points of those who live and work in the same municipality and a 0.013 decline in live-and-work share, the same direction as the earned income tax and statistically significant. The school-district property rate is also positive, similar to the income tax effect above, and is only statistically significant for the live and work outcome with an estimate of 0.017. The county rate is null in all four outcomes but also positive signed. The property tax effects are also all smaller than the EIT estimates, possibly indicating that property taxes increases induce smaller migration effects, while EIT adjustments do not. The four fiscal spending shares are individually indistinguishable from zero on all outcomes. Section 5.2 returns to

the property coefficients, where the same pattern appears in the decomposition.

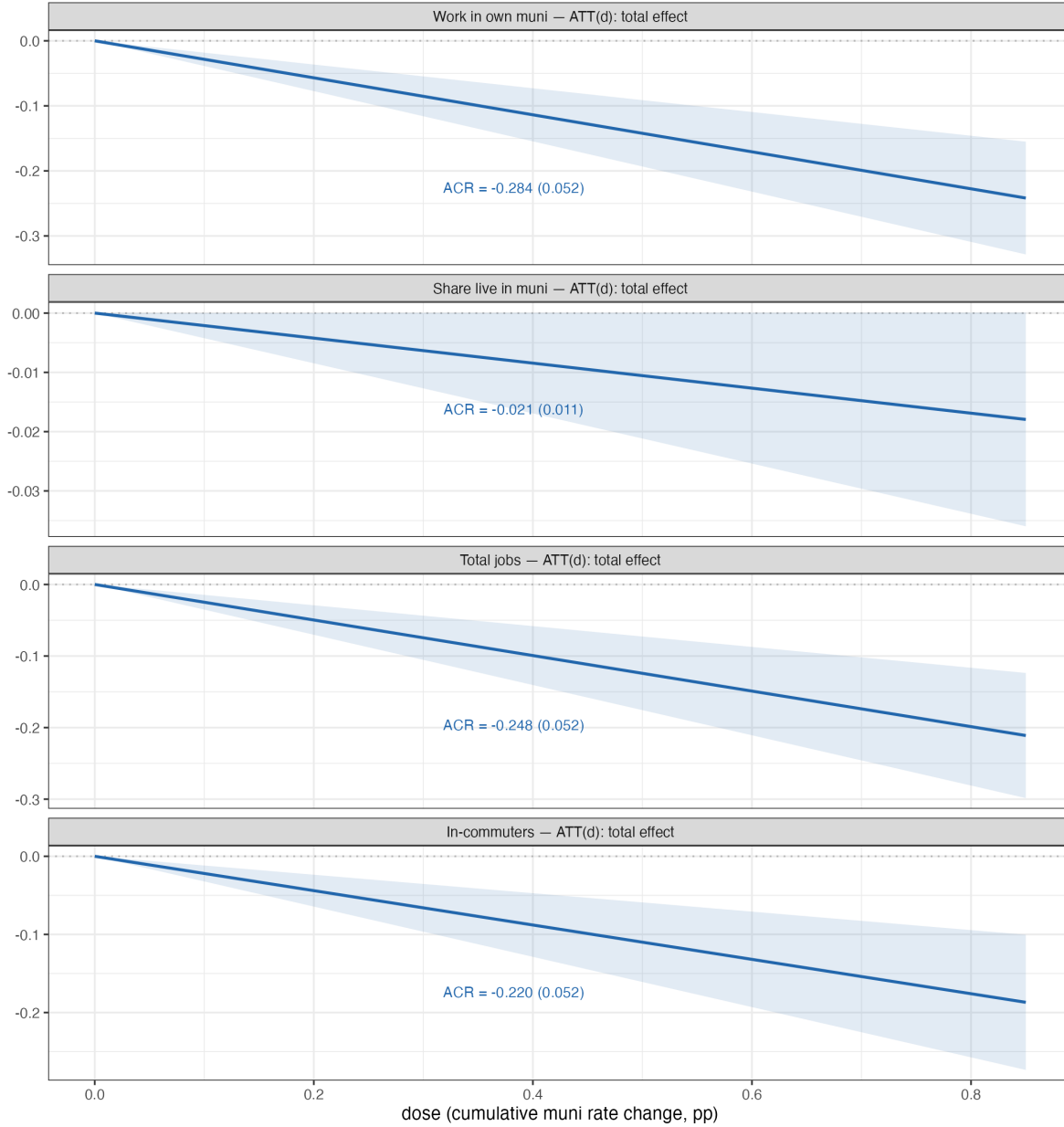
Figure 5 draws the dose response and the ACR reported in Table 3 above. Under the linear specification $ATT(d)$ is a straight downward sloping line from the origin whose slope is the ACR, the marginal effect $ACRT(d)$ equals that slope at every dose, and the three quantities are one number. The shaded band is the 95% interval, which widens with the dose because extrapolating a slope compounds its uncertainty. Whether the response really is straight over this range is a fair question, and Appendix A answers it by re-estimating the same specification with a quadratic and with a natural spline. Both stay within a standard error of the line, so I do not expect the linear restriction to be biasing the coefficient, though the flexible fits do suggest the marginal effect flattens at the largest increases, which would mean municipalities making the biggest increases have already taxed away the last marginal mover and those remaining will not move at any rate in this range. These estimates are from the pooled sample. Below in Section 5.2 I estimate the tax increases by utilizing the binding mechanism.

Figure 5: Dose-response, linear specification: pooled Increase.

Dose-response, linear specification — Pooled Increase

Linear specification, net of SD rate changes.

$ATT(d) = ACR \times d$ with a 95% band. No ACRT panel, since under the linear model the marginal effect is constant and equal to the ACR.



Notes: $ATT(d) = ACR \times d$ from the headline specification, with a delta-method 95% band. The marginal effect is constant under the linear model, so there is no ACRT panel.

5.2 Decomposition Results

Table 4 reports the estimates from the binding decomposition model. This model is slightly different than the pooled specification estimated above. This model instead breaks down the pooled model by whether the municipal or school district rate increase was non-binding, becomes binding, or was already binding.

Table 4: Municipality-level dose-response: muni and SD resident-rate cells (joint regression)

Cell	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
<i>Muni resident rate (τ^M)</i>				
Inc-bind	-0.058*** (0.019)	-0.537*** (0.141)	-0.345*** (0.119)	-0.423*** (0.124)
Inc-becomes	-0.015 (0.011)	-0.250*** (0.048)	-0.212*** (0.056)	-0.233*** (0.055)
Inc-nobind	-0.028 (0.027)	-0.110 (0.151)	0.145 (0.225)	0.114 (0.196)
<i>School district rate (τ^{SD})</i>				
Inc-bind	0.006 (0.007)	0.171 (0.105)	0.170* (0.097)	0.177* (0.094)
Inc-becomes	0.007 (0.007)	0.086 (0.095)	0.087 (0.089)	0.097 (0.089)
<i>Fiscal controls</i>				
police-services share	0.002 (0.007)	0.024 (0.092)	-0.091 (0.096)	-0.090 (0.093)
streets share	0.001 (0.003)	0.051 (0.041)	0.019 (0.041)	0.021 (0.041)
debt-service share	0.004 (0.005)	-0.045 (0.054)	-0.026 (0.061)	-0.018 (0.059)
intergovernmental-revenue share	0.008** (0.004)	0.040 (0.045)	0.026 (0.050)	0.034 (0.049)
muni property eff. rate (pp)	-0.013** (0.005)	-0.213*** (0.051)	-0.110** (0.045)	-0.129*** (0.044)
SD property eff. rate (pp)	0.001 (0.001)	0.017** (0.009)	0.009 (0.009)	0.010 (0.009)
county property eff. rate (pp)	0.001 (0.002)	0.031 (0.024)	0.011 (0.022)	0.013 (0.022)
<i>Diagnostics</i>				
<i>N</i>	187,815	171,451	187,807	187,815
<i>R</i> ²	0.826	0.963	0.959	0.961
Within <i>R</i> ²	0.001	0.002	0.001	0.001

Notes: Stacked-DiD; muni×stack and year×stack FE; SE clustered at muni. Realized-dose pre-trend tests for every cell and outcome are reported in the pre-trend battery. Both rate-setters enter the same joint regression; the table reports both sets of cell×dose coefficients with the shared fiscal controls.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Starting with the municipal tax effects and the cell where the home stack was already binding. I find that a one point increase reduces the count of workers who live and work in the municipality by 0.537 log points, or 41.6 percent, and reduces their share of local jobs by 5.84 percentage points. Both are significant at one percent and both are the outcomes the paper leans on, because they describe residents rather than changes to the local labor market. The two commuter-side outcomes move in the same direction but are generally smaller, in-commuters by 0.345 log points and total jobs by 0.423. While these estimates show the labor market is declining while the EIT is raised, the ordering of the effect sizes match what is expected. When the municipal resident rate effects the tax bill of someone who lives and works in the municipality and never reaches the bill of someone who commutes in, the outcome that mixes the two (logged in-commuters) should land between them, and it does. Another implications of the decline in both jobs and those who live and work in the municipality is that the share falls because the two counts do not fall together. The share of those that live and work in the community is defined as the number of those that work and live in a municipality divided by the total jobs of the municipality. Because the resident workforce drops by 41.6 percent per point and total jobs by 34.5 percent, the number of residents leave faster than the job base contracts. A general decline in local employment would, in theory, move both counts by similar amounts and the share would be near zero. However, the EIT is affecting only resident workers, so their decline is disproportionate to the general decline of the labor market.

Like in the pooled model, a one point increase is larger than any rate increase observed in the data. The largest cumulative increase is 0.50 points while an average of 0.24 points over five years after their first increase, the resident workforce falls by 12.2 percent, the live-and-work share by 1.42 percentage points, total jobs by 9.8 percent and in-commuters by 8.0 percent.

The increase-becomes cell, where the increase pushed the home stack above the work rate, estimates about half the effect of the inc-bind cell. The resident workforce falls by

0.250 log points per point, or 22.1 percent, with in-commuters at 0.212 and total jobs at 0.233, all significant at one percent. The live-and-work share is the exception and is not statistically significant. This effect highlights that the tax increase when it becomes binding has yet to have the disproportionate impact on those that live and work in the municipality. The EIT increase may drive out those that live and work given the effect on the share that lives and works in the municipality is negative, I cannot be fully confident on the magnitude. These municipalities raised rates by more than Inc-bind, 0.38 points on average against 0.24, so at their own average increase the resident workforce falls by 9.2 percent and total jobs by 8.5 percent.

The third cell (inc-nobind) is not statistically significant on any outcome. Where the rate rose but the home stock stayed below the work rate, the resident workforce outcomes are both negative while the two commuter outcomes are positive. While the labor market in the Inc-nobind cities may be growing relative to the control group, the effect of the municipal rate increase, may still induce migration because of the sign on the resident workforce outcomes, even though I cannot be confident in the size of the estimate. Lastly, the effect size declines as the tax becomes less binding, possibly indicating that as a greater share of tax gets placed on income workers are more likely to move.

The school-district cells tell a similar story to the pooled specification. Every coefficient is positive, none reaches five percent significance level, and only total jobs and in-commuters in the binding cell reach ten percent significance. The effects are discussed in Section 5.1, where the same pattern appears in the pooled specification.

The property-tax controls show that a one percentage point higher municipal effective property rate is associated with 0.213 fewer log points of resident workforce, the same direction as the earned income tax. The magnitude is roughly two and a half times smaller than the binding-cell EIT response per point of rate.

The other two layers run the other way and shrink as the taxing jurisdiction grows. The school-district rate is statistically significant for the logged work in municipality outcome and

finds an effect of 0.017 for each percentage point increase, while the county rate has a positive 0.031 effect while not being statistically significant across outcomes. A positive coefficient on school-district and county property taxes predictable through the Tiebout model, since they both are the level at which a local tax most visibly buys a services households sort toward and property taxes are the dominant source of school funding in Pennsylvania. Lastly, property taxes capitalize into house prices, so a household buying into a high-rate municipality is paying into services from their perpetual higher burden experienced. Earned income taxes do not capitalize the same way, since they fall on an individuals labor, and relocation is one of the few margins available for avoiding them.

Figure 6 draws the dose response for the Inc-bind cell reported in Table 4. The line is the estimating equation rather than an illustration of it: under the linear specification $ATT(d)$ is $ACR \times d$, a ray from the origin whose slope is the coefficient in the table, and the marginal effect is that same number at every dose, which is why no second panel appears.

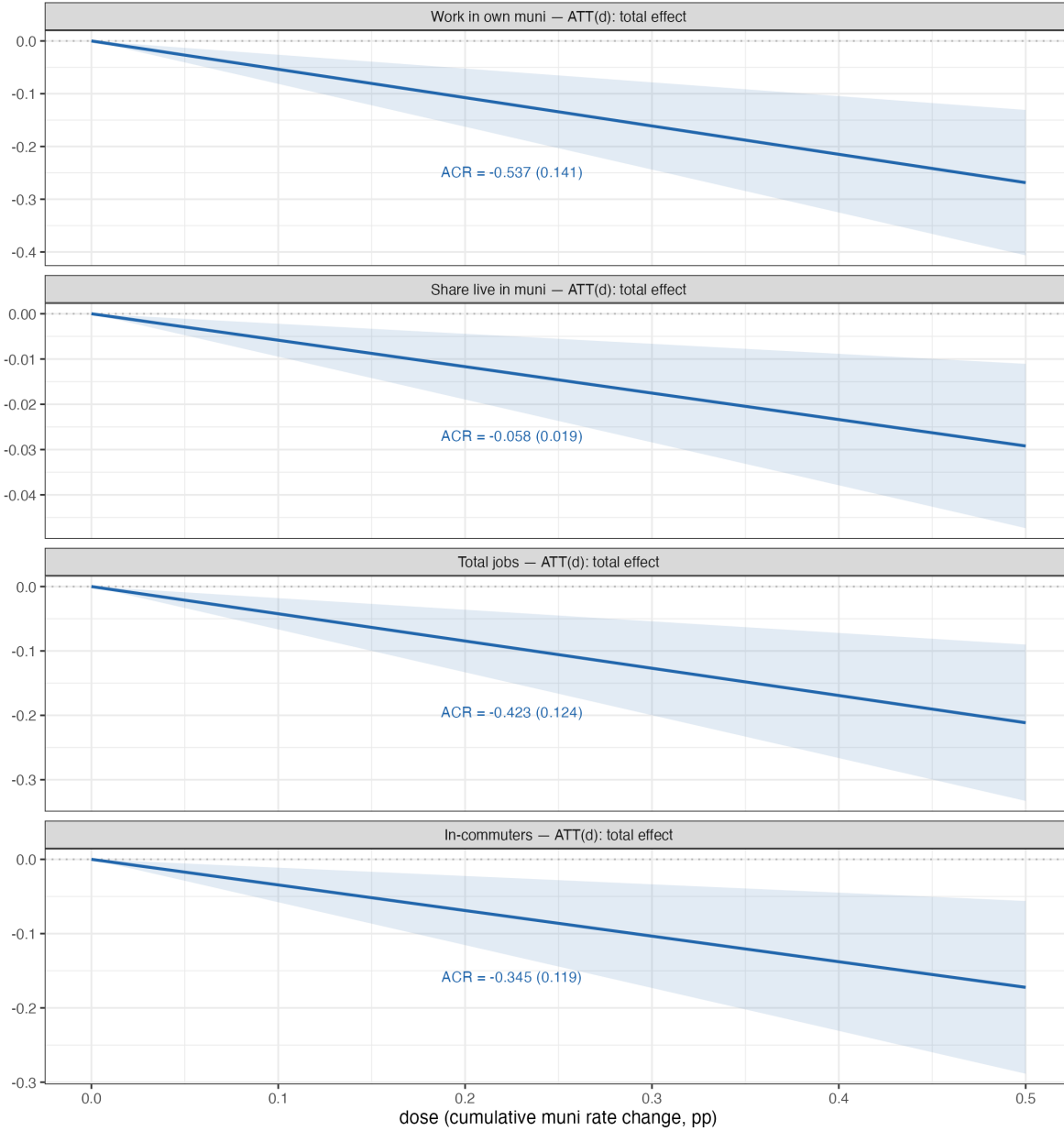
What the figure adds to the table is the band. It widens with the dose because the uncertainty in a slope compounds as it is extrapolated, so the effect at half a percentage point is far less pinned down than the effect at a tenth. That matters here, since half a point is the largest cumulative increase this cell contains and the per-point coefficient is being read at a dose no municipality reached. Appendix A re-estimates the same specification with a quadratic and with a natural spline in the dose.

Figure 6: Dose-response, linear specification: Inc-bind cell.

Dose-response, linear specification — Inc-bind

Linear specification, net of SD rate changes.

$ATT(d) = ACR \times d$ with a 95% band. No ACRT panel, since under the linear model the marginal effect is constant and equal to the ACR.



Notes: $ATT(d) = ACR \times d$ from the headline specification, with a delta-method 95% band. The marginal effect is constant under the linear model, so there is no ACRT panel.

Figure 7 does the same for the Inc-becomes cell, where the increase newly pushed the home stack above the work rate. Its line is shallower, at -0.250 against -0.537 , and its municipalities took larger increases, so the two cells arrive at similar totals from opposite

directions. Its band is also much tighter, because thirty-nine municipalities pin a slope down better than twenty-four. The Inc-nobind cell has no counterpart figure, since its estimates are not distinguishable from zero and its line would be flat.

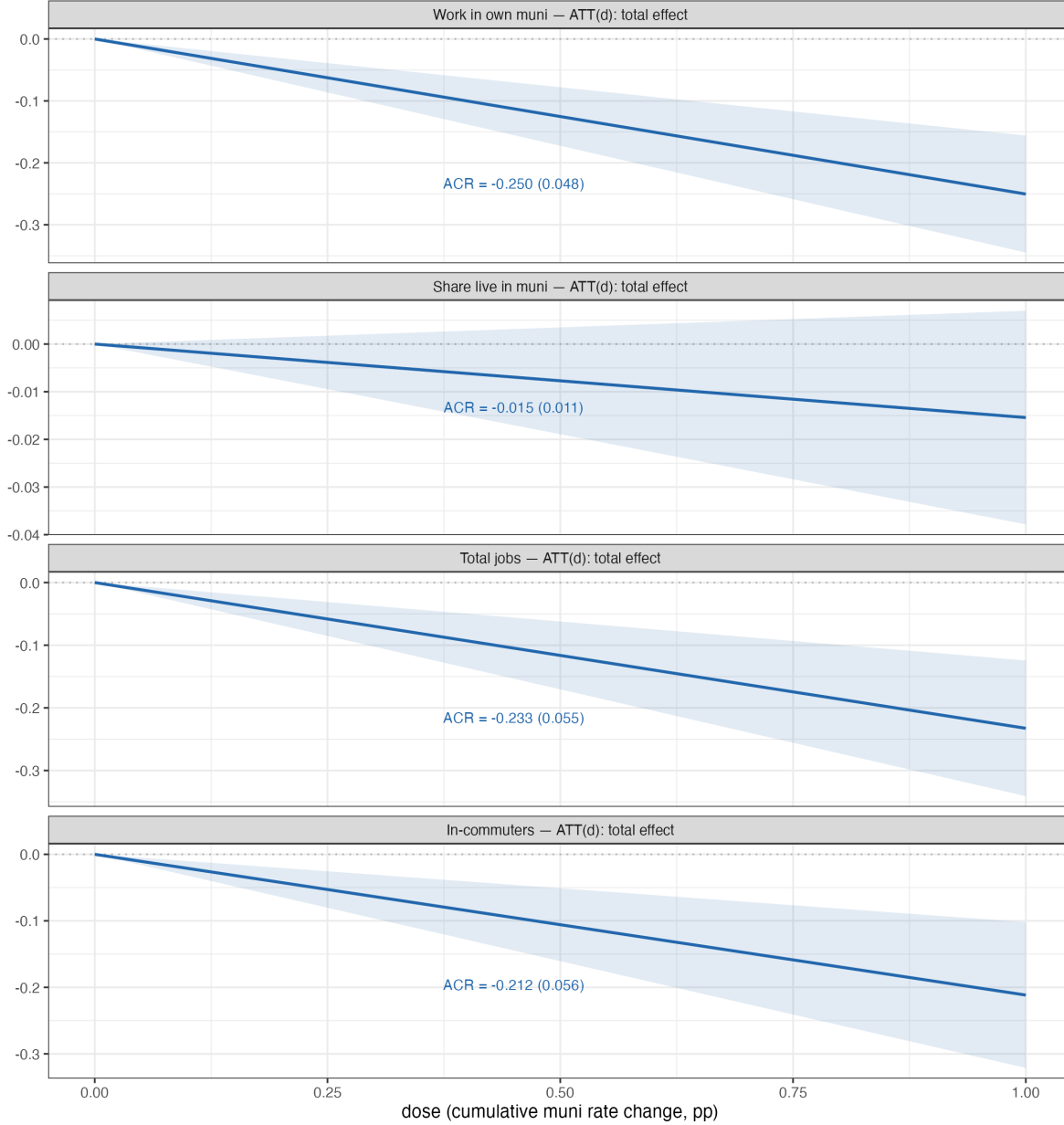
Whether the response is really a straight line over this range is a separate question, and Appendix A takes it up by re-estimating both cells with a quadratic and with a natural spline.

Figure 7: Dose-response, linear specification: Inc-becomes cell.

Dose-response, linear specification — Inc-becomes

Linear specification, net of SD rate changes.

$ATT(d) = ACR \times d$ with a 95% band. No ACRT panel, since under the linear model the marginal effect is constant and equal to the ACR.



Notes: $ATT(d) = ACR \times d$ from the headline specification, with a delta-method 95% band. The marginal effect is constant under the linear model, so there is no ACRT panel.

5.3 Event-Study Results

This section describes the event study results. The event study was modeled using the realized dose over the sample period. I elected for this event study design over a binary because it is able to answer whether the response appears when the rate changes rather than before it and whether municipalities that eventually make large increases were already moving differently from those that make small ones. Figure 8 plots the event-time paths, and the full set of leads and lags behind it is reported cell by cell in Tables 12 through 15.

The leads are the coefficients at event times -5 through -2 , each measuring how far apart large-dose and small-dose municipalities already were in that year relative to the year before the rate changed. The joint Wald test asks whether all four leads are zero at the same time. A large p -value means the test cannot detect a pre-existing divergence, which is the desired outcome, and a small p -value indicates cities are moving on different paths prior to treatment.

Start with the count of workers who live and work in the municipality. In Inc-bind it is flat across the pre-event window and then declines steadily once the rate changes, reaching -0.789 per percentage point by the fifth year. Its four leads average -0.116 and the joint Wald returns $p = 0.053$, which is a narrow pass. Inc-becomes shows the same post-event shape at roughly a third the size, -0.242 by the fifth year, but its leads average $+0.045$ and the Wald rejects at $p = 0.001$. Inc-nobind is not shown in the figure and fails the pre-trends test with a Wald value below 0.001 . The pooled specification, which mixes all three, rejects at $p = 0.029$. Only the cell where the bill was already moving passes the test.

The share of local jobs held by residents behaves differently. Here Inc-bind is the cell that rejects, at $p = 0.005$, even though its post-event path is the largest and most precisely estimated of the four. Inc-becomes does not reject ($p = 0.293$), nor does the pooled specification ($p = 0.259$), while Inc-nobind again rejects below 0.001 .

The two commuter-side outcomes present a different set of results. In-commuters and total jobs Inc-bind is the cleanest cell in the table, with Wald values of 0.775 and 0.783 , while

Inc-becomes rejects on both at $p < 0.001$ and Inc-nobind at 0.005 and 0.001. Inc-bind's leads on these outcomes are positive and imprecise and should be noted that in figure 8 the commuter-side shows a general decline over the entire sample from -5 to $+5$.

Across the four outcomes the Wald test does not provide a single verdict on whether the strong parallel trends assumption is satisfied or not. Visually, the pre-trends do look to satisfy the assumption as well as looking at the pre-period coefficients where only a few are statistically significant across outcome variables. In Section 6.4 runs the joint Wald alongside a linear pre-slope, which tests the alternative that actually threatens the design, namely treated and control municipalities drifting apart at a steady rate before the tax changes. The joint Wald rejects on any departure in any shape, including one stray year. Across the sixteen cell-by-outcome rows the Wald rejects on nine, and in the slope test on only three. Where both the Wald and the linear pre-slope reject, the estimate should be interpreted cautiously and not causally, which is the case for in-commuters and total jobs in Inc-becomes.

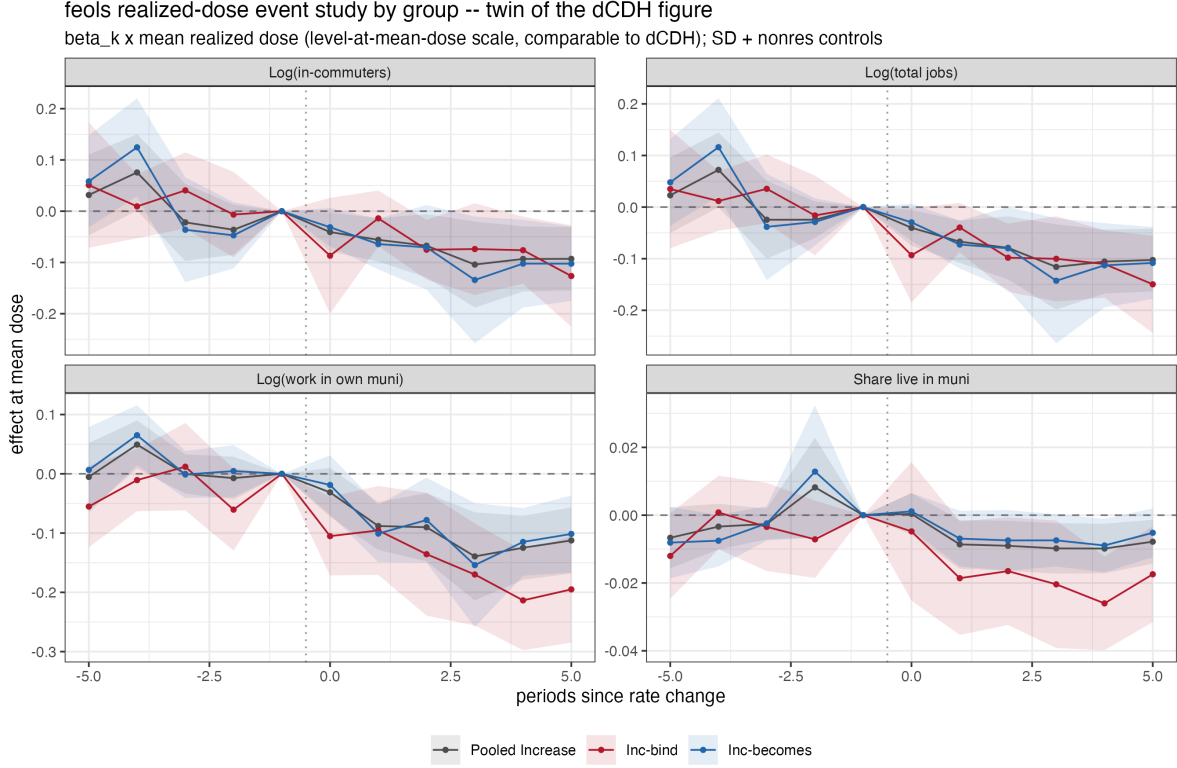
For Inc-bind on the resident workforce no lead is individually significant, the linear pre-slope is $+0.024$ and indistinguishable from zero ($p = 0.47$), and the joint Wald sits at $p = 0.053$. Extrapolated forward its pre-period predicts a mild increase rather than the decline that follows. What the battery establishes is which estimates survive being looked at from several directions, and the Inc-bind effect on the resident workforce is the one that does.

The de Chaisemartin and D'Haultfœuille estimator in Section 6.1 supports these results on both counts. Its effect estimates line up with the ones above once the two are put on the same scale, and its placebo detects no pre-trend on any of the four outcomes, with p -values between 0.23 and 0.69. That is a friendlier outcome than the one this section reports, and the reason is that the two tests ask different questions and therefore estimate differently. The placebo compares municipalities that switch against those that have not switched yet, which is average parallel trends and is what a binary design requires. The realized-dose test tags each municipality with the rate change it will eventually take and asks whether the

size of that change predicts the pre-event path, which is strong parallel trends and is what a continuous treatment requires (Callaway et al., 2024). Strong parallel trends is stricter than average parallel trends, and is the one required for this design. Section 6.1 takes the comparison up in full.

Taken together the results support three claims. First, municipalities that raise the earned income tax lose resident workers, and at the average increase the loss is on the order of nine to twelve percent depending on whether the increase is already binding (inc-bind). Second, the response increases in size as the tax liability increases which is a stronger predictor than the rate increase itself, for example, Inc-nobind raised rates by more than any other cell and shows no statistically significant effect. Third, the identification is secure for one cell and one outcome and weaker elsewhere. The Inc-bind effect on the count of workers who live and work in the municipality survives every diagnostic applied to it, though the joint Wald only narrowly. The remaining cells and outcomes carry pre-trends the design cannot rule out, and I read them further validating evidence.

Figure 8: Realized-dose event study by group (pooled / Inc-bind / Inc-becomes), level-at-mean-dose scale.



6 Robustness

This section summarizes the robustness checks. It carries on as follows. Section 6.1 re-estimates the response with an estimator that assigns no negative weights and imposes no functional form on the dose. Section 6.2 splits the sample at the Act 32 enforcement break. Section 6.3 drops the municipality that was under state-supervised financial recovery. Section 6.4 runs three pre-trend tests on every cell and outcome, and Section 6.5 assigns placebo event dates that precede the real ones. Full tables sit in Appendix D, and the pooled counterpart of each check is in Appendix C.

6.1 de Chaisemartin and D’Haultfœuille Estimator

Each headline estimate is assumed to be the ACR, and because I rely on a continuous stacked difference design, two issues may cause biased estimates. First, the design uses two-way fixed effects which in a continuous model potentially generates negative weights. Second, the design is linear in nature, and could potentially bias the estimate across dosages as the linear fit may over or underestimate the relationship of the dose to the outcome variable as shown in Figures 10, 11 and 12. The linear-in-dose stacked specification recovers it cleanly only if the weights are non-negative and the dose-response is not strongly curved (de Chaisemartin et al., 2024). I test both concerns by re-estimating the muni-side increase response on the same sample with the non-negative-weight continuous-treatment estimator of Chaisemartin et al. (2025) (dCDH estimator).

Table 5: Cumulative effect at $k = +5$ by group, de Chaisemartin and D’Haultfœuille estimator

Group	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Pooled Increase	-0.005	-0.092**	-0.111**	-0.115**
	(0.008)	(0.045)	(0.051)	(0.047)
Avg total effect per pp ($\approx$ ACR)	-0.019	-0.197	-0.183	-0.206
Joint placebo p	0.693	0.258	0.234	0.316
N (at $k = +5$)	11,279	9,952	11,279	11,279
Switchers	33	32	33	33
Inc-bind	0.003	-0.052	-0.095*	-0.090**
	(0.014)	(0.075)	(0.052)	(0.045)
Avg total effect per pp ($\approx$ ACR)	-0.012	-0.092	-0.139	-0.155
Joint placebo p	0.000	0.553	0.218	0.296
N (at $k = +5$)	9,428	8,288	9,428	9,428
Switchers	17	17	17	17
Inc-becomes	-0.013**	-0.158***	-0.139	-0.154*
	(0.007)	(0.058)	(0.094)	(0.093)
Avg total effect per pp ($\approx$ ACR)	-0.022	-0.278	-0.218	-0.243
Joint placebo p	0.000	0.445	0.037	0.005
N (at $k = +5$)	9,527	7,049	9,527	9,527
Switchers	14	13	14	14

Notes: Effects are levels at the realized dose, not per-point slopes, so compare them to the linear ACR times the group’s mean dose. The per-pp row is already dose-normalized. Larger placebo p means no detectable pre-trend. Estimated with the `continuous` option and analytical standard errors, which the package documentation notes can be liberal, so the standard errors are a lower bound. Controls are partialled out within baseline-dose groups, so no control coefficients exist to report. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (normal approximation).

Table 5 reports each group in a set of rows. The bold top row is the cumulative effect at $k = +5$ with its standard error beneath, which is interpreted as the total change in the outcome five years after the rate change. Because that is interpreted as a level change rather than a rate change it is not comparable to the headline coefficients. The row below it, the average total effect per pp, is the estimator’s own dose-normalized figure, and that one is comparable to headline coefficients. I will be focusing on the effect in this row and acknowledging the statistical significance in the first row. The three rows beneath report

the joint placebo test, where larger values mean treated, and control municipalities were on similar paths before treatment. The last two rows are the number of observations and treated municipalities (switchers).

In the pooled specification the two estimators are nearly identical on outcomes. Reading the Avg total effect per pp row against Table 3, the share of local jobs held by residents is -0.019 against -0.021 , and the count of workers who live and work in the municipality is -0.197 against -0.284 . In-commuters are -0.183 against -0.220 and total jobs -0.206 against -0.248 . The dCDH estimator is the smaller of the two on every outcome, indicating a slight change when adjusting for linearity and negative weights.

Looking at the binding decomposition rows the estimates show the same sign as the headline results but the magnitudes change. The Inc-bind estimates diverge on every outcome. On the share of workers who live and work in the municipality I find -0.012 against the headline -0.058 , on the count of those workers -0.092 against -0.537 , on in-commuters -0.139 against -0.345 , and on total jobs -0.155 against -0.423 . The dCDH estimator is three to six times smaller in every column. Inc-becomes behaves the opposite way and lands close on all four, at -0.022 against -0.015 on the share, -0.278 against -0.250 on the resident workforce, -0.218 against -0.212 on in-commuters and -0.243 against -0.233 on total jobs, slightly larger each time rather than smaller.

I do not have a clear understanding of why the inc-bind estimates are vastly different. Some possible reasons are that each model is run on its own subsample of never-changers, the headline is stacked, with repeated never-changer copied and stack-interacted fixed effects, while the dCDH estimator runs on the unstacked panel. The headline coefficient is an average post-treatment level shift per unit of rate, while the figure here accumulates effects across periods before normalizing by the treatment received. Additionally, the two do not use the same number of units, since Inc-bind is identified off 17 municipalities (switchers) here against 24 municipalities in the headline.

The joint placebo test in the table is this estimator's own pre-trend check. It estimates

how the periods before the treatment were either similar or moving independently, and tests whether they are jointly zero. A large p -value means it cannot detect a divergence and is the favourable outcome. Inc-bind passes on the three count outcomes, at 0.553 on the resident workforce, 0.218 on in-commuters and 0.296 on total jobs, and fails on the share at 0.000. Inc-becomes passes on the resident workforce at 0.445 and fails on the other three, at 0.000 on the share, 0.037 on in-commuters and 0.005 on total jobs.

Figure 9 plots the event studies that connect to the p -values summarized. The two commuter outcomes decline steadily after the rate change in every group and most steeply for Inc-becomes, which reaches -0.208 on in-commuters and -0.223 on total jobs by the fifth year. The pre-event periods, while not being statistically significant, are also in a general decline. Inc-becomes is also the group whose pre-event path diverges from the other groups. It runs positive through the first three leads and then dips at $t - 2$ and $t - 1$, to -0.164 and -0.107 on in-commuters. That dip is what the placebo rejects at 0.037 and 0.005.

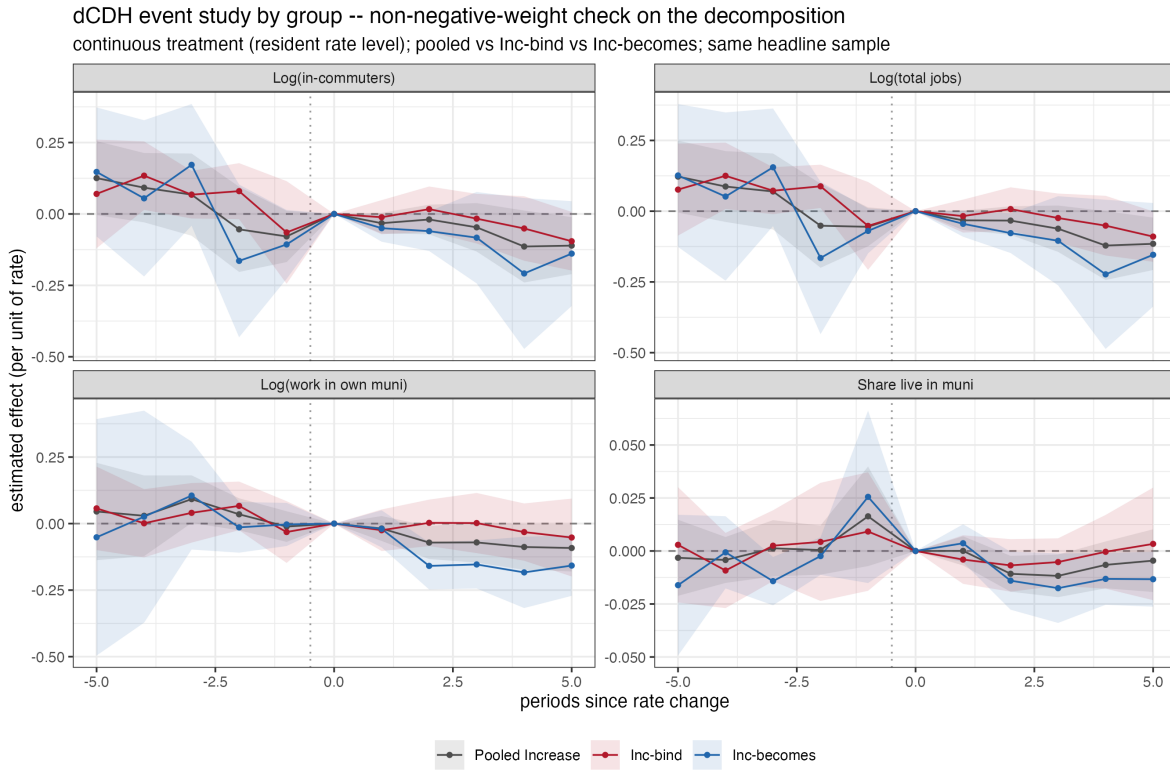
The resident-side outcomes behave differently. Pre-event paths on the count of workers who live and work in the municipality are flat in all three groups, wandering near zero without direction, and the placebo clears every one of them at 0.553, 0.445 and 0.258.

The share is the exception, where all three groups sit near zero through the first four leads and then jump in the period immediately before the rate changes, by $+0.009$ for Inc-bind, $+0.026$ for Inc-becomes and $+0.016$ pooled. The jump in the share outcome and the variations in pre-periods is likely causing the rejection at 0.000. The pooled group shows the same jump and clears at 0.693, but is generally smoother across pre-periods than Inc-becomes.

These outcomes and the realized-dose test of Section 5.3 disagree at the pooled level and agree on the rows that matter most. Both reject on the Inc-bind share, at 0.000 here against 0.005 there. Both reject on the Inc-becomes commuter outcomes, at 0.037 and 0.005 here against 0.000 and 0.000 there. Two tests built on different comparisons and different parallel-trends assumptions flag the same three rows. Where they do not agree is the resident

outcomes, which the realized-dose test rejects for Inc-becomes and this one does not, and which this test clears for Inc-bind more than the realized-dose test does. Only one of the two agreements is visible in the figure. The Inc-becomes commuter panels climb through the early leads and then swing negative, which is a trend an eye can find. The Inc-bind share is flat on the page and rejects on precision alone.

Figure 9: Event-study paths by group, de Chaisemartin and D’Haultfoeulle estimator



Notes: Effects are levels at the realized dose, not per percentage point. Reference period is $k = 0$.

Because the dCDH estimator corrects for negative weighting that may occur in the two-way fixed effects model and the pooled coefficients in the stacked DiD model are similar to one another, the estimates indicate that the linear specification is not seriously distorted by the weighting problem that heterogeneous effects can create in a two-way fixed effects regression. The linear results are generally smaller of the two, so the linear coefficient is likely not biased by producing the wrong sign or order of magnitude. Also the estimator imposes no functional form on the dose, and given the coefficients the linear approximation

is not losing much to possible decreasing returns over higher changes in the rate. Its joint placebo test detects no pre-trend on any of the four outcomes at the pooled level, with p -values between 0.23 and 0.69.

6.2 Pre/Post Act 32 Decomposition

This section splits the specification at the Act-32 enforcement break (cohort first event year < 2012 vs ≥ 2012) and tests whether the binding-cell effect strengthens under consolidated employer withholding (see Appendix D.1).

The results show that the effect actually weakens in the post-2012 period. Inc-bind is essentially unchanged on three of the four outcomes, at -0.544 before the break against -0.531 after on the resident workforce, -0.337 against -0.362 on in-commuters and -0.432 against -0.415 on total jobs. Each of those differences sits well inside either standard error and every one of the six coefficients is significant at five percent. The largest deviation is the live-and-work share, which halves from -0.068 to -0.043 while remaining significant on both sides.

Inc-becomes attenuates modestly across the board, from -0.296 to -0.222 on the resident workforce, -0.218 to -0.209 on in-commuters and -0.276 to -0.203 on total jobs. The share outcome is the one coefficient that changes sign, from -0.043 (significant to 10%) pre-2012 to $+0.004$ after and is not statistically significant. Inc-nobind has no pre-2012 estimate at all. Eleven of the municipalities that would be estimated for this cell fall on pre-2012 side of the break and their dose term is collinear with the stack fixed effects on the thin pre-enforcement subsample, so the estimator drops it.

The hypothesized effect was that consolidated withholding should have made the tax harder to avoid and that the response should be larger in the post-period. The model shows the opposite. One reason this may occur is that Act 32 changed collection rather than tax liability, such that a household already paying its home rate faced the same bill before and after and had the same reason to move.

Table 6: Act 32 split, pre-enforcement cohorts (first event before 2012): full specification

Outcome	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Inc-bind	-0.068** (0.029)	-0.544*** (0.201)	-0.337** (0.166)	-0.432** (0.173)
Inc-becomes	-0.043* (0.023)	-0.296*** (0.107)	-0.218** (0.109)	-0.276** (0.119)
<i>Fiscal controls</i>				
police-services share	0.002 (0.007)	0.021 (0.093)	-0.094 (0.097)	-0.093 (0.094)
streets share	0.002 (0.003)	0.051 (0.041)	0.019 (0.041)	0.021 (0.041)
debt-service share	0.004 (0.005)	-0.046 (0.054)	-0.026 (0.061)	-0.018 (0.059)
intergovernmental-revenue share	0.009** (0.004)	0.041 (0.045)	0.026 (0.050)	0.035 (0.050)
muni property eff. rate (pp)	-0.013** (0.005)	-0.214*** (0.051)	-0.110** (0.045)	-0.130*** (0.044)
SD property eff. rate (pp)	0.001 (0.001)	0.017** (0.009)	0.009 (0.009)	0.010 (0.009)
county property eff. rate (pp)	0.001 (0.002)	0.031 (0.024)	0.011 (0.022)	0.012 (0.022)
<i>N</i>	187,663	171,302	187,655	187,663
<i>R</i> ²	0.826	0.963	0.959	0.961
Within <i>R</i> ²	0.001	0.002	0.001	0.001
Treated pre-period baseline	—	—	—	—

Notes: Stacked-DiD; muni×stack and year×stack FE; SE clustered at muni. Inc-nobind is absent pre-2012: 11 municipalities are in the cell, but their dose term is collinear with the stack fixed effects on the thin pre-enforcement subsample and is dropped by the estimator. Rows omitted (no estimable coefficients): Inc-nobind. SE in parens. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Act 32 split, post-enforcement cohorts (first event in/after 2012): full specification

Outcome	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Inc-bind	-0.043*** (0.005)	-0.531*** (0.162)	-0.362** (0.150)	-0.415*** (0.147)
Inc-becomes	0.004 (0.010)	-0.222*** (0.047)	-0.209*** (0.056)	-0.203*** (0.052)
Inc-nobind	-0.028 (0.027)	-0.110 (0.152)	0.145 (0.224)	0.114 (0.196)
<i>Fiscal controls</i>				
police-services share	0.002 (0.007)	0.019 (0.093)	-0.095 (0.097)	-0.094 (0.094)
streets share	0.001 (0.003)	0.052 (0.041)	0.019 (0.041)	0.022 (0.041)
debt-service share	0.004 (0.005)	-0.047 (0.054)	-0.026 (0.061)	-0.019 (0.059)
intergovernmental-revenue share	0.008** (0.004)	0.041 (0.045)	0.026 (0.050)	0.034 (0.050)
muni property eff. rate (pp)	-0.013** (0.005)	-0.213*** (0.051)	-0.109** (0.045)	-0.129*** (0.044)
SD property eff. rate (pp)	0.001 (0.001)	0.017** (0.009)	0.009 (0.009)	0.010 (0.009)
county property eff. rate (pp)	0.001 (0.002)	0.031 (0.024)	0.011 (0.022)	0.012 (0.022)
<i>N</i>	187,479	171,136	187,471	187,479
<i>R</i> ²	0.826	0.963	0.959	0.961
Within <i>R</i> ²	0.001	0.002	0.001	0.001
Treated pre-period baseline	—	—	—	—

Notes: Stacked-DiD; muni×stack and year×stack FE; SE clustered at muni. SE in parens. * $p < 0.10$,

** $p < 0.05$, *** $p < 0.01$.

6.3 Act 47 Distressed-Municipality Exclusion

Act 47 is the legal designation that lets a municipality exceed the 0.5% statutory municipality-EIT cap meaning there may be a selection confounder (Municipalities Financial Recovery Act (Act 47), 1987). An Act 47 designation may come from a period of population decline, economic decline, changes in the labor force, and service cuts. Matched against DCED's

own register of distressed determinations and rescissions, exactly 1 municipalities has ever carried the designation, Hazleton, and was designated in 2017, seven years after the 2010 rate increase that places it in the cell (“Act 47 Financial Distress”, n.d.). Hazleton is an Inc-bind municipality. Other historically designated cities do are not in the sample or other cells because the reversal test in Section 4.4 removes them, and of the sixteen register entries absent from the estimation panel seven are removed on because the nonresident rate is missing and dropped from the sample.

Table 8 re-estimates the Inc-bind coefficients without Hazleton. The resident workforce outcome variable falls from -0.537 to -0.439 , in-commuters from -0.345 to -0.283 , total jobs from -0.423 to -0.345 , and the live-and-work share from -0.058 to -0.048 . Every outcome loses a noticeable magnitude, but every one stays statistically significant at five percent. This model says that one municipality of twenty-four carries a large share in the effect size for Inc-bind cell, but still accurately predicted sizes.

Table 8: Inc-bind effect excluding Hazleton, the one Act 47 designated municipality

Cell	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
<i>Muni resident rate (τ^M)</i>				
Inc-bind	-0.048** (0.019)	-0.439*** (0.134)	-0.283** (0.126)	-0.345*** (0.123)
Inc-becomes	-0.015 (0.011)	-0.250*** (0.048)	-0.212*** (0.056)	-0.233*** (0.055)
Inc-nobind	-0.028 (0.027)	-0.110 (0.151)	0.145 (0.225)	0.114 (0.196)
<i>School district rate (τ^{SD})</i>				
Inc-bind	0.006 (0.007)	0.171 (0.105)	0.170* (0.097)	0.177* (0.094)
Inc-becomes	0.007 (0.007)	0.086 (0.095)	0.087 (0.089)	0.097 (0.089)
<i>Fiscal controls</i>				
police-services share	0.002 (0.007)	0.024 (0.092)	-0.091 (0.096)	-0.090 (0.093)
streets share	0.001 (0.003)	0.051 (0.041)	0.019 (0.041)	0.021 (0.041)
debt-service share	0.004 (0.005)	-0.045 (0.054)	-0.026 (0.061)	-0.018 (0.059)
intergovernmental-revenue share	0.008** (0.004)	0.040 (0.045)	0.026 (0.050)	0.034 (0.049)
muni property eff. rate (pp)	-0.012** (0.005)	-0.213*** (0.051)	-0.110** (0.045)	-0.129*** (0.044)
SD property eff. rate (pp)	0.001 (0.001)	0.017** (0.009)	0.009 (0.009)	0.010 (0.009)
county property eff. rate (pp)	0.001 (0.002)	0.031 (0.024)	0.011 (0.022)	0.013 (0.022)
<i>Diagnostics</i>				
<i>N</i>	187,805	171,441	187,797	187,805
<i>R</i> ²	0.826	0.963	0.959	0.961
Within <i>R</i> ²	0.001	0.002	0.001	0.001

Notes: Stacked-DiD; muni×stack and year×stack FE; SE clustered at muni. Leave-one-out on the Inc-bind cell, dropping Hazleton. Identical specification and dose source to the headline table, so the cells other than Inc-bind reproduce it. Both rate-setters enter the same joint regression; the table reports both sets of cell×dose coefficients with the shared fiscal controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Pre-Trend Battery

Table 9 reports three pre-trend tests, each sensitive to a different kind of failure. The joint Wald tests all four pre-event leads against zero simultaneously. It rejects on any departure of any shape, which makes it the most demanding of the three and also the one most easily upset by a single stray coefficient or by a poorly estimated covariance matrix on a thin cohort. A clean result is a large p -value.

The count of individually significant leads asks a narrower question: is any one pre-event coefficient distinguishable from zero at five percent. A clean result is zero of four.

The linear pre-slope fits a single trend through the pre-event window and tests its slope on one degree of freedom. It is insensitive to an isolated spike and identifies treated and control municipalities drifting apart at a steady rate before the treatment event. A clean result is a slope indistinguishable from zero.

The three tests do not all agree and drawing from the distinction that the disagreement supports in Section 5.3. The joint Wald rejects on nine of the sixteen cell-by-outcome rows, the linear pre-slope on three, and only five rows carry any individually significant lead. Two rows fail on both the Wald and the slope, in-commuters ($p = 0.0003$ joint, $p = 0.001$ slope) and total jobs ($p = 0.0004$ and $p = 0.001$), both in Inc-becomes. Of the remaining seven Wald rejections, four are the whole of Inc-nobind, where no lead is individually significant and the slope rejects on none of the four outcomes, and the other three are the Inc-bind share, the Inc-becomes resident workforce and the pooled resident workforce, each with a flat slope.

The table is not a certificate and is not built as one. It is a sorting device, and what it sorts to the top is Inc-bind, which clears all three tests on three of its four outcomes and fails the joint Wald only on the share. What it sorts to the bottom is the pair of Inc-becomes commuter outcomes, which fail all three and should be read as descriptive rather than causal. Inc-nobind sits outside the sort, failing the joint Wald on everything while passing the other two on everything, which is what thirteen municipalities produce and is

why the cell is treated as uninformative rather than clean. The short version for a reader is that the design supports a causal reading for one cell and corroborating evidence elsewhere, which is a narrower claim than the count of significant coefficients in Table 4 would suggest.

Table 9: Pre-trend battery: three tests per binding cell

Cell	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Inc-bind (24 munis)				
Joint Wald p (leads)	0.005	0.053	0.775	0.782
Leads significant at 5% (of 4)	0	0	0	0
Linear pre-slope p	0.223	0.472	0.350	0.404
Robust clean	yes	yes	yes	yes
Inc-becomes (39 munis)				
Joint Wald p (leads)	0.293	0.001	0.000	0.000
Leads significant at 5% (of 4)	0	1	1	1
Linear pre-slope p	0.155	0.157	0.001	0.001
Robust clean	yes	no	no	no
Inc-nobind (13 munis)				
Joint Wald p (leads)	0.000	0.000	0.005	0.001
Leads significant at 5% (of 4)	0	0	0	0
Linear pre-slope p	0.149	0.380	0.251	0.265
Robust clean	yes	yes	yes	yes
Pooled Increase (76 munis)				
Joint Wald p (leads)	0.259	0.029	0.060	0.103
Leads significant at 5% (of 4)	0	1	1	0
Linear pre-slope p	0.253	0.189	0.043	0.051
Robust clean	yes	no	no	no

Notes: All three tests are run on the realized-dose leads, and larger p means no detectable pre-trend.

Leads significant counts individually significant pre-event coefficients. Robust clean requires no significant lead and a flat slope. The joint Wald over-rejects on thin cohorts and does not drive the verdict.

The pooled row is shown for comparison.

6.5 Fake-Timing Placebo

Every test in Section 6.4 tests the same sample of pre-event coefficients. This check is different in that each treated municipality is assigned a placebo event three, four, and five years before its real one, and only data from before the real event occurs is allowed in the

estimation. The design then estimates an effect where no rate changed at all, meaning the coefficients should be null. A failure here means the specification finds an effect out of a pre-period where there should not be one, which is a different issue from treated and control municipalities moving in different direction in the pre-period.

The Inc-bind cell passes on both live-and-work and share of residents working in a municipality. Its live-and-work coefficients are $+0.053$, $+0.071$ and -0.033 across the three shifts, none distinguishable from zero, and its share coefficients are 0.000 , $+0.020$ and $+0.003$.

Inc-becomes does not pass. Its live-and-work placebo is -0.173 at the three-year shift ($p = 0.013$) and -0.179 at five years ($p = 0.055$). Read together with that cell's failure on both pre-trend tests in Section 6.4, its coefficients carry the weakest causal warrant in the paper. The non-binding cell's share fails at the four- and five-year shifts (-0.074 and -0.065).

The three panels overlap heavily and are consistency checks rather than independent tests, and eligibility requires two placebo pre-years inside the panel, so the municipality count falls as the shift lengthens.

Table 10: Fake-timing placebos: events shifted 3–5 years early, pre-event data only

Cell	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
<i>Placebo event at $c - 3$ (37 eligible munis)</i>				
Inc-bind	0.000 (0.016)	0.053 (0.074)	-0.035 (0.123)	-0.041 (0.113)
Inc-becomes	0.017 (0.010)	-0.173** (0.070)	-0.334*** (0.102)	-0.313*** (0.094)
Inc-nobind	-0.109 (0.066)	-0.265 (0.199)	0.997 (0.953)	0.877 (0.881)
<i>Placebo event at $c - 4$ (31 eligible munis)</i>				
Inc-bind	0.020 (0.015)	0.071 (0.047)	-0.167 (0.187)	-0.144 (0.170)
Inc-becomes	0.009 (0.009)	-0.122 (0.094)	-0.226** (0.089)	-0.213** (0.086)
Inc-nobind	-0.074** (0.037)	-0.086 (0.359)	0.776 (0.832)	0.695 (0.792)
<i>Placebo event at $c - 5$ (27 eligible munis)</i>				
Inc-bind	0.003 (0.014)	-0.033 (0.067)	-0.100 (0.138)	-0.100 (0.122)
Inc-becomes	0.005 (0.009)	-0.179* (0.093)	-0.243*** (0.087)	-0.237*** (0.084)
Inc-nobind	-0.065** (0.031)	-0.108 (0.363)	0.578 (0.760)	0.507 (0.727)

Notes: Each treated municipality is assigned a placebo event 3, 4 or 5 years before its real one and only data preceding the real event enter, so every coefficient should be null. The three panels share municipalities and overlapping windows, so they are consistency checks rather than independent tests, and eligibility varies with the shift. SE in parens. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Discussion

In the results, I find that general tax increases drive migration out of municipalities, but that result is mostly due to the binding decomposition. A municipality that raises a rate where

the tax is already binding loses 0.537 log points of resident workforce per percentage point. One whose increase now pushes the home stack above the work rate (becomes binding) loses 0.250 log points of resident workforce per percentage point. One whose rate rises while the tax bill stays below the work rates finds no statistically significant effect. Only the already binding and becomes binding changed what anyone owed, and are the only two to show a response.

The comparison with the existing literature is directly comparable, because no two of these papers share an outcome variable nor the same methodology. Yang and Heim (2017) study county-level aggregate taxable income in Indiana, the total income reported on state returns filed in a county, and conclude that taxpayers are unresponsive to small changes in local rates. Yushkov (2025) are the most similar to my design but use county-to-county migration counts from IRS records, and finds inflows and outflows that are negatively signed but not statistically significant in most specifications. While the designs and outcome variables may not be directly comparable, the useful contrast between them is the sign of their estimate and statistical significance and not necessarily the magnitude of the effect. The binding-cell estimates are negative, which is the direction the mechanism predicts, and in most cases statistically significant at the one percent level on the resident-side outcomes, where Yushkov (2025) and Yang and Heim (2017) work returns coefficients that are not statistically significant. Two possible reasons for the differences are that: (1) LODES counts the workers who live in a municipality, so the outcome variable is the population of interest that the resident rate actually falls on rather than an aggregate. (2) I utilize a continuous stacked difference-in-differences design that uses the size of each rate change in the model, which keeps municipalities with small and large increases in the same sample without condensing the additional variation that provides into a single treatment. My pooled specification, which is the closest in design to theirs, returns -0.284 against -0.537 in the binding cell. The results show that averaging municipalities whose bills moved under binding or non-binding changes the pooled outcome.

The Pennsylvania setting has been studied before for compliance rather than for sorting. Bagchi (2022) shows that Act 32’s consolidated withholding raised compliance by about fourteen percent. That finding does sit well with the findings in Section 6.2, which returns -0.544 before the break and -0.531 after. If enforcement brought a larger share of residents into actually paying the tax, a rate increase would mean a higher coefficient after the break compared to the pre-period. However, before Act 32 employers were not generally required to withhold the EIT, except employees who lived and worked in the same municipality (Bagchi, 2022). The residents whose compliance Act 32 changed were mostly those working somewhere other than where they lived, and Bagchi (2022) finds the compliance gains smallest in districts where the most residents lived and worked in the same place. The enforcement landed largely outside the live and work outcome variable, so finding relatively little effect is not unsurprising. The two papers also do not share the same unit, where his estimates are at the school district and mine at the municipality, so the comparison is directional but completely comparable. What the split establishes is that the effect is not an artifact of the enforcement regime, and not that enforcement is irrelevant.

Livas and Jacob (2026) also study earned income taxes in Pennsylvania but specifically on Philadelphia’s wage tax and the implications on suburbanization. Philadelphia’s levy sits outside the Act 511 system, the tax rate is much higher, and is excluded from my sample. However, our research finds commonality in that what is estimated is the response to a change in local income taxes in the same state and the resulting migration due to that tax increase. The two papers are complements in that they estimate responses to different instruments at different scales, and the estimate in my paper is assessing what the effect for an average Pennsylvania municipality can expect from a typical rate increase.

While there is not a copious literature on local income taxes in the United States, there are many studies focusing on Switzerland. Comparing against the Swiss research the comparison is more difficult and an extrapolation between outcomes. Basten et al. (2017) identifies the effect of tax differences between municipal boundaries and finds a decrease in rents and high

willingness to pay as incomes increase, while Schmidheiny (2006) finds that higher income households are more likely to move to low-income municipalities, and Schaltegger et al. (2011) finds similar results from the income composition of communities across the canton of Zurich. One major difference between my paper and the Swiss papers is that Swiss municipal income taxes are progressive, so a rate difference gives high-income households a stronger reason to move than low-income ones, whereas my research looks at a flat tax. Pennsylvania’s EIT, being flat does not offer the same incentive by income, and LODES worker counts do not identify income at a level that would let me test for one. However, the research may permit extrapolation in that it is possible the individuals in my sample are wealthier earners and that as taxes increase higher income individuals move out but as the dosage reaches a certain level that may stall and the last marginal individual willing to move is taxed out. Agrawal and Foremny (2019) supports this argument using data from Spain. After Spain’s 2011 reform let regions set their own top marginal rates, they estimate an elasticity of the number of top taxpayers and find that a 1 percent increase in a region’s rate relative to others raises the probability of moving there by 1.7 percentage points. The same progressivity caveat applies, since the Spanish variation is in a top marginal rate and mine is a flat rate paid at every income.

There are three limitations worth discussing. First, the estimates are not equally well identified. The binding cell passes the fake-timing placebo at all three shifts and shows no linear pre-trend, while the inc-becomes cell fails both pre-trend tests and the placebo on its commuter outcomes. Second, the sample is restricted to municipalities whose rate paths do not reverse, which removes Pittsburgh, Allentown, Reading, and Johnstown among others, so the estimates describe monotone rate-setters rather than the full state of municipalities. Lastly, in-commuters make up 89 percent of total jobs and sit in the denominator of the live-and-work share, and the municipal resident rate never touches an in-commuter’s bill. I utilized the commuter side outcomes as a measurement of general labor market health.

The property-tax coefficients in Section 5.2 offer a different perspective on taxation in

Pennsylvania municipalities. A percentage point of municipal effective property tax is associated with a 0.213 log-point decline in the count of workers who live and work in the municipality, against 0.537 for a binding EIT increase, so the income tax moves roughly two and a half times as many workers per point of rate. That ordering is an association rather than a causal estimate, since the property rates enter as controls and receive none of the decomposition, event-study, and placebo tests the EIT estimates do. An earned income tax falls on labor income, so the base and the resident are the same taxable commodity and one leaves as the other increases. A property tax falls in part on land, which cannot relocate, and it capitalizes into house prices, so a share of the burden settles on the owner at the time of the increase. The difference is about revenue rather than efficiency. A municipality that sorts households away from its income tax loses the base as those individuals move, while one that sorts households away from a land tax keeps the land, the price adjusts, and the revenue stays. Pennsylvania levies its property tax on land and structures together, so the 0.213 blends an immobile base with a reproducible one. This is largely commentary on the need for revenue, but pulling from a base that is mobile proves that other forms of taxation, such as increasing them on inelastic goods, may have a higher return to a municipality than on taxes on labor. Agrawal and Foremny (2019) report that the mechanical revenue increase from higher rates exceeds the revenue lost to net outmigration, so an income tax shrinks its base without giving back the whole gain. What this paper can say is that the instrument a municipality chooses determines whether raising the rate erodes the thing being taxed at all.

In terms of policy, the statutory increase produces a large response, a smaller one, or none at all depending on where a municipality's rate sits relative to the work-side rate its residents face. A municipality raising a rate that already binds is choosing a different policy from one raising a rate that does not, and the revenue and residential consequences of the two are not the same.

8 Conclusion

This paper estimates the movement of workers when a local income tax rises. In Pennsylvania, workers move when the increase is clearly changing their tax bill. This is identifiable by using the piecewise structure of the Earned Income Tax to separate rate changes that alter the tax liability from rate changes that do not, I find that municipalities raising a binding rate lose resident workers at 0.537 log points per percentage point, that municipalities whose increase newly becomes binding lose 0.250, and that municipalities whose rate rises without affecting the tax bill show no statistically significant response. At the average increase, Inc-bind loses about twelve percent of the workers who live and work there.

The identification in this paper is a novel approach to estimating the causal effect of local income tax increases on worker migration. While some cells and outcomes do not remain perfectly identified given the plethora of diagnostics run in Section 6, the results generally show a statistically significant and predictable response. The evidence supports the response that tax liability experienced rather than statutory rates, through the law determine whose bill moves is important to consider versus simply averaging over.

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A Flexible Dose Specifications

The headline enters the dose linearly, this appendix drops that restriction and estimates the same model with a quadratic and with a spline in the dose.

The three specifications differ in the dose term and nothing else:

$$Y_{it} = \alpha_i + \gamma_t + \beta d_{it} + \mathbf{X}'_{it}\phi + \varepsilon_{it} \quad (7)$$

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 d_{it} + \beta_2 d_{it}^2 + \mathbf{X}'_{it}\phi + \varepsilon_{it} \quad (8)$$

$$Y_{it} = \alpha_i + \gamma_t + B(d_{it})'b + \mathbf{X}'_{it}\phi + \varepsilon_{it} \quad (9)$$

d_{it} is the cumulative change in the municipal rate and $\mathbf{X}_{it}$ the fiscal controls. $B(\cdot)$ in Equation (9) is a natural cubic spline with interior knots at the terciles of the positive treated doses and boundary knots at zero and the largest observed dose. Each dose term is interacted with the binding cell, as in the headline. The sample, controls, school-district co-treatment, stacked municipality and year fixed effects and clustering are the headline's, and all three share one complete-cases set, so N is the same in every column.

One difference from the headline is deliberate. The headline regressor is the rate level and these are the cumulative change. For a linear term the two are the same once municipality fixed effects absorb the baseline rate, which is why Equation (7) reproduces the headline exactly in Table 11. They are not the same for a nonlinear term, because $d^2 = b_i^2 + 2b_i c_{it} + c_{it}^2$ leaves a baseline-by-change term the fixed effects cannot remove. Fitting Equation (8) in levels flips the sign of both coefficients, so the choice matters. The change is what the paper's dose-response describes.

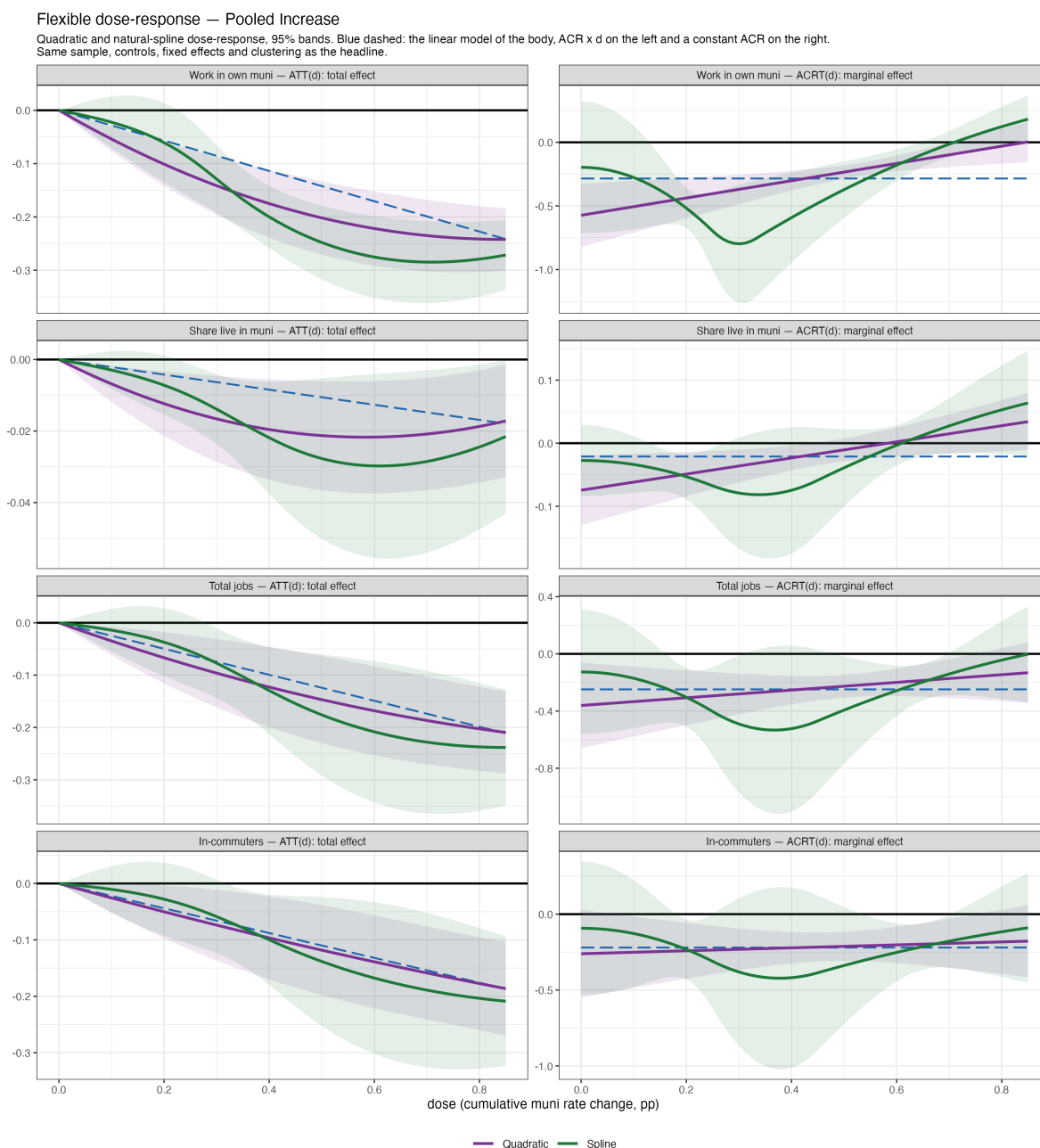
Table 11: ACR under three dose specifications

	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Pooled Increase				
Linear (headline)	-0.021 [*] (0.011)	-0.284 ^{***} (0.052)	-0.220 ^{***} (0.052)	-0.248 ^{***} (0.052)
Quadratic	-0.030 ^{***} (0.011)	-0.336 ^{***} (0.049)	-0.226 ^{***} (0.060)	-0.267 ^{***} (0.061)
Natural spline	-0.040 ^{**} (0.017)	-0.415 ^{***} (0.054)	-0.262 ^{***} (0.090)	-0.318 ^{***} (0.091)
Inc-bind				
Linear (headline)	-0.058 ^{***} (0.019)	-0.537 ^{***} (0.141)	-0.345 ^{***} (0.119)	-0.423 ^{***} (0.124)
Quadratic	-0.062 ^{***} (0.020)	-0.507 ^{***} (0.129)	-0.288 [*] (0.165)	-0.371 ^{**} (0.154)
Natural spline	-0.081 ^{***} (0.031)	-0.376 ^{**} (0.156)	-0.021 (0.239)	-0.127 (0.205)
Inc-becomes				
Linear (headline)	-0.015 (0.011)	-0.250 ^{***} (0.048)	-0.212 ^{***} (0.056)	-0.233 ^{***} (0.055)
Quadratic	-0.023 (0.014)	-0.295 ^{***} (0.047)	-0.213 ^{***} (0.068)	-0.244 ^{***} (0.069)
Natural spline	-0.028 [*] (0.016)	-0.336 ^{***} (0.043)	-0.226 ^{***} (0.077)	-0.264 ^{***} (0.079)
Inc-nobind				
Linear (headline)	-0.028 (0.027)	-0.110 (0.151)	0.145 (0.225)	0.114 (0.196)
Quadratic	-0.890 ^{***} (0.025)	-3.937 (2.425)	6.294 ^{**} (1.338)	5.289 ^{***} (1.343)
Natural spline	—	—	—	—

Notes: Each row is the ACR, the average of $ACRT(d)$ weighted by $\omega(d)$, which puts one weight on each treated municipality at the dose it took. The linear row is the headline coefficient. Mean doses: Pooled Increase 0.350 (77 munis), Inc-bind 0.243 (24 munis), Inc-becomes 0.400 (39 munis), Inc-nobind 0.391 (14 munis). The quadratic and spline rows enter the dose as the cumulative change rather than the rate level, which is the same as the headline for a linear term but not for a nonlinear one. Sample, controls, school-district co-treatment, fixed effects and clustering are the headline's, $N = 171,451$. Inc-nobind's 14 treated municipalities cannot identify a spline. SE in parens. ^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

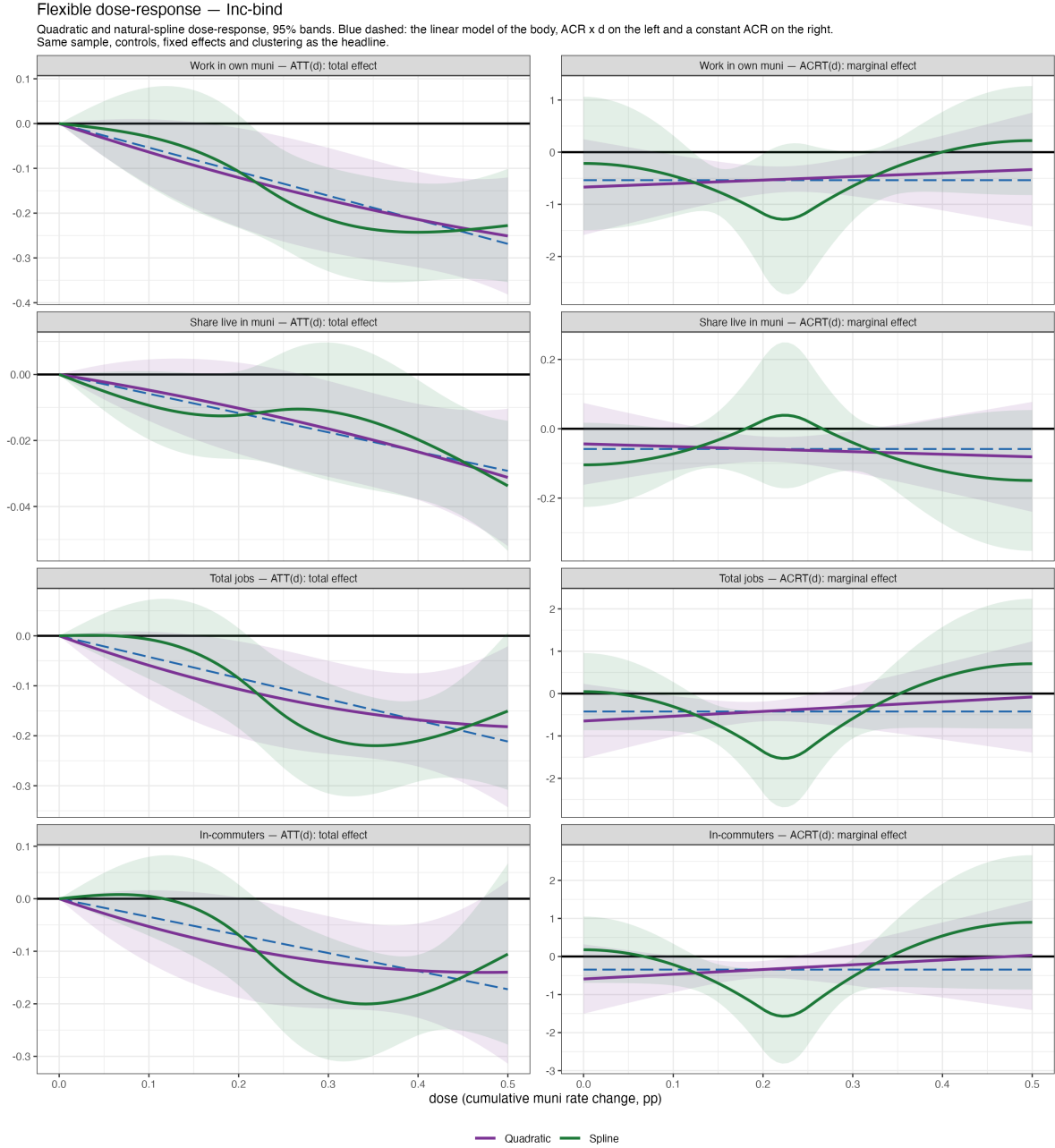
Figures 10 to 12 draw the two flexible fits against the linear model of the body. The left panels are $ATT(d)$ and the right panels the marginal effect $ACRT(d)$, which is where the three specifications differ. $ACRT(d)$ is constant by construction under the linearity, a straight tilt under the quadratic, and varies under the spline.

Figure 10: Flexible dose-response: pooled Increase.



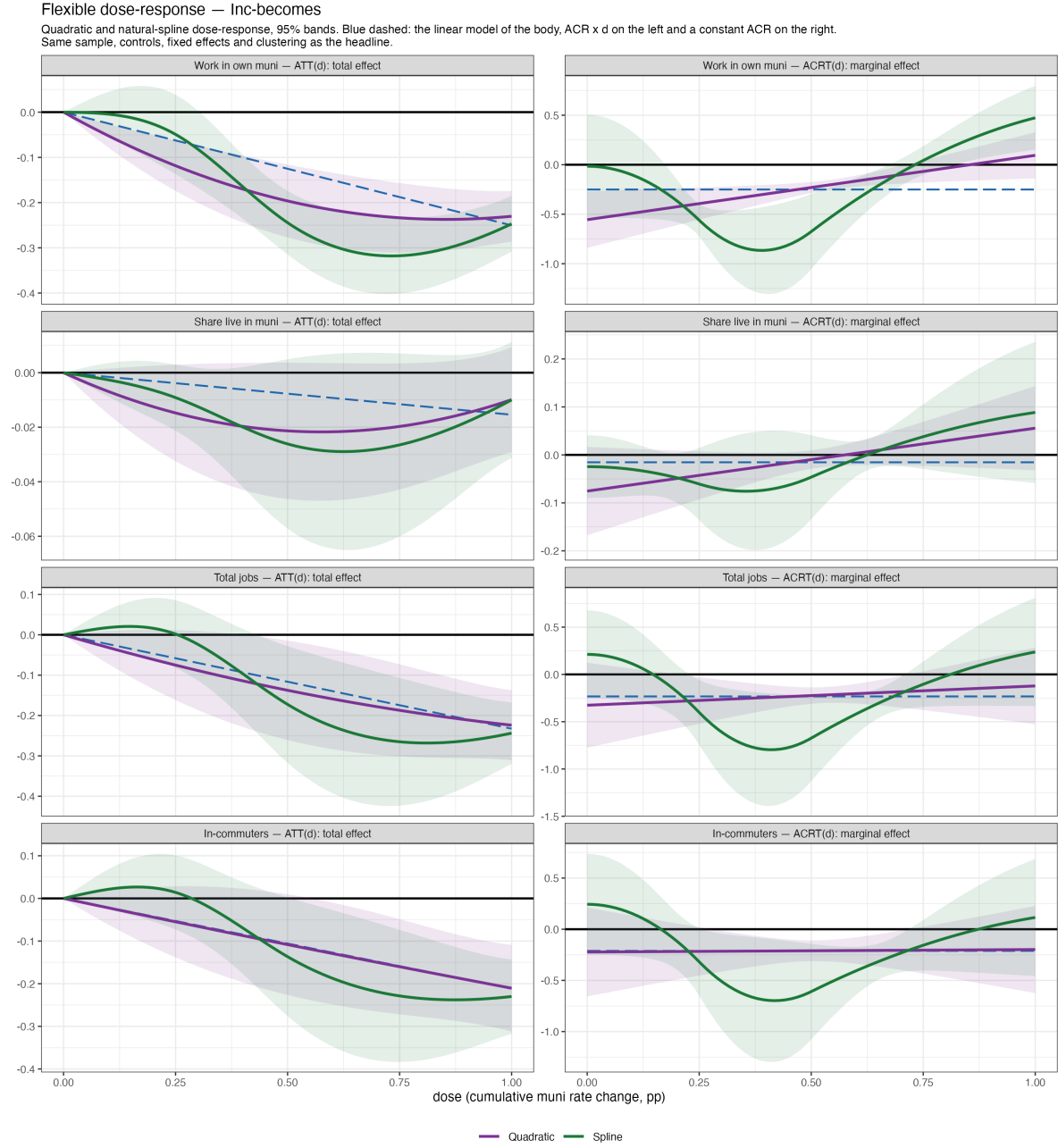
Notes: Quadratic and natural-spline dose-response with 95% bands. Blue dashed is the linear model of Figure 5. Sample, controls, fixed effects and clustering are the headline's.

Figure 11: Flexible dose-response: Inc-bind cell.



Notes: As Figure 10, against the linear model of Figure 6. The spline bands are wide because three basis columns are being fit on twenty-four treated municipalities.

Figure 12: Flexible dose-response: Inc-becomes cell.



Notes: As Figure 10, against the linear model of Figure 7. This is the cell whose curvature is statistically detectable.

B Realized-Dose Event Studies, Full Set

Table 12: Realized-dose event study, full coefficients: Pooled Increase

Event time	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
$k = -5$	-0.019 (0.013)	-0.015 (0.081)	0.088 (0.112)	0.064 (0.104)
$k = -4$	-0.009 (0.010)	0.138** (0.058)	0.211** (0.106)	0.201* (0.103)
$k = -3$	-0.007 (0.006)	-0.001 (0.047)	-0.061 (0.107)	-0.069 (0.107)
$k = -2$	0.023 (0.021)	-0.020 (0.051)	-0.101 (0.069)	-0.068 (0.050)
$k = 0$	0.001 (0.009)	-0.088 (0.059)	-0.114** (0.054)	-0.112** (0.049)
$k = 1$	-0.024** (0.010)	-0.246*** (0.057)	-0.156*** (0.060)	-0.187*** (0.058)
$k = 2$	-0.025** (0.010)	-0.252*** (0.082)	-0.188** (0.088)	-0.220** (0.087)
$k = 3$	-0.027*** (0.011)	-0.389*** (0.106)	-0.291** (0.119)	-0.324*** (0.117)
$k = 4$	-0.027*** (0.010)	-0.349*** (0.077)	-0.260*** (0.090)	-0.295*** (0.088)
$k = 5$	-0.022** (0.009)	-0.314*** (0.079)	-0.260*** (0.088)	-0.286*** (0.086)
N	189,168	172,795	189,160	189,168
R^2	0.825	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001
Joint Wald p (leads)	0.259	0.029	0.060	0.103

Notes: Per-pp realized-dose coefficient at each event time; $k = -1$ omitted as the reference. SE clustered by municipality. Joint Wald tests the four pre-event leads against zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Realized-dose event study, full coefficients: Inc-bind cell

Event time	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
$k = -5$	-0.049* (0.026)	-0.223 (0.142)	0.206 (0.252)	0.142 (0.236)
$k = -4$	0.003 (0.022)	-0.043 (0.108)	0.038 (0.128)	0.048 (0.119)
$k = -3$	-0.014 (0.027)	0.048 (0.152)	0.164 (0.153)	0.143 (0.137)
$k = -2$	-0.029 (0.023)	-0.245* (0.142)	-0.027 (0.172)	-0.066 (0.159)
$k = 0$	-0.019 (0.042)	-0.425*** (0.137)	-0.351 (0.233)	-0.377** (0.189)
$k = 1$	-0.075** (0.035)	-0.386** (0.154)	-0.056 (0.112)	-0.160 (0.098)
$k = 2$	-0.067** (0.033)	-0.548** (0.214)	-0.304** (0.120)	-0.397*** (0.136)
$k = 3$	-0.082** (0.039)	-0.687*** (0.177)	-0.299 (0.184)	-0.405** (0.170)
$k = 4$	-0.105*** (0.028)	-0.864*** (0.173)	-0.308** (0.135)	-0.446*** (0.134)
$k = 5$	-0.070** (0.029)	-0.789*** (0.185)	-0.512** (0.204)	-0.605*** (0.194)
N	188,846	172,497	188,838	188,846
Municipalities	1708	1687	1708	1708
R^2	0.825	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001
Joint Wald p (leads)	0.005	0.053	0.775	0.782

Notes: Per-pp realized-dose coefficient at each event time; $k = -1$ omitted as the reference. SE clustered by municipality. Joint Wald tests the four pre-event leads against zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: Realized-dose event study, full coefficients: Inc-becomes cell

Event time	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
$k = -5$	-0.019 (0.013)	0.016 (0.088)	0.139 (0.109)	0.114 (0.102)
$k = -4$	-0.018* (0.009)	0.155** (0.061)	0.297** (0.116)	0.276** (0.115)
$k = -3$	-0.006 (0.006)	-0.003 (0.048)	-0.086 (0.124)	-0.091 (0.125)
$k = -2$	0.031 (0.024)	0.011 (0.053)	-0.112 (0.078)	-0.068 (0.055)
$k = 0$	0.003 (0.007)	-0.045 (0.060)	-0.075 (0.045)	-0.070 (0.044)
$k = 1$	-0.016* (0.010)	-0.240*** (0.058)	-0.153** (0.060)	-0.173*** (0.056)
$k = 2$	-0.018 (0.011)	-0.186** (0.087)	-0.168* (0.100)	-0.190** (0.097)
$k = 3$	-0.018* (0.009)	-0.367*** (0.127)	-0.320** (0.150)	-0.340** (0.147)
$k = 4$	-0.021** (0.010)	-0.274*** (0.069)	-0.244** (0.104)	-0.269*** (0.099)
$k = 5$	-0.012 (0.009)	-0.242*** (0.079)	-0.244*** (0.088)	-0.258*** (0.084)
N	188,930	172,568	188,922	188,930
Municipalities	1723	1702	1723	1723
R^2	0.825	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001
Joint Wald p (leads)	0.293	0.001	0.000	0.000

Notes: Per-pp realized-dose coefficient at each event time; $k = -1$ omitted as the reference. SE clustered by municipality. Joint Wald tests the four pre-event leads against zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15: Realized-dose event study, full coefficients: Inc-nobind cell (placebo)

Event time	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
$k = -5$	0.087 (0.056)	0.146 (0.236)	-0.931 (0.811)	-0.836 (0.753)
$k = -4$	0.111 (0.080)	0.378 (0.281)	-0.922 (0.830)	-0.797 (0.745)
$k = -3$	0.009 (0.014)	0.163 (0.372)	0.060 (0.096)	0.070 (0.110)
$k = -2$	0.004 (0.005)	0.141 (0.099)	0.101 (0.089)	0.105 (0.089)
$k = 0$	0.008 (0.011)	0.166 (0.182)	0.021 (0.036)	0.029 (0.033)
$k = 1$	-0.014 (0.015)	0.122 (0.193)	-0.053 (0.187)	-0.070 (0.180)
$k = 2$	-0.020 (0.015)	-0.104 (0.132)	0.074 (0.104)	0.052 (0.095)
$k = 3$	-0.014 (0.014)	0.034 (0.140)	0.078 (0.097)	0.064 (0.089)
$k = 4$	0.027 (0.020)	0.081 (0.205)	-0.030 (0.096)	0.004 (0.088)
$k = 5$	-0.011 (0.015)	0.110 (0.168)	0.211 (0.222)	0.201 (0.218)
N	188,750	172,390	188,742	188,750
Municipalities	1698	1676	1698	1698
R^2	0.825	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001
Joint Wald p (leads)	0.000	0.000	0.005	0.001

Notes: Per-pp realized-dose coefficient at each event time; $k = -1$ omitted as the reference. SE clustered by municipality. Joint Wald tests the four pre-event leads against zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Pooled realized-dose event study — municipality level.

Direction	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Inc (post)	-0.021 (0.004)	-0.273 (0.032)	-0.211 (0.035)	-0.237 (0.034)
Inc (pre)	-0.003 (0.007)	0.025 (0.030)	0.034 (0.050)	0.032 (0.047)
Inc Wald p	0.2589	0.0290	0.0599	0.1031
<i>Diagnostics</i>				
N	189,168	172,795	189,160	189,168
R^2	0.825	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001
Dep. SD	0.076	1.923	1.881	1.895

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \dots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of $H_0: \beta_k = 0$ across pre-event k . $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 17: Realized-dose event study by binding-state cell — municipality level.

Cell	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Inc-bind (post)	-0.070 (0.014)	-0.617 (0.072)	-0.305 (0.070)	-0.398 (0.064)
Inc-bind (pre)	-0.022 (0.012)	-0.116 (0.068)	0.095 (0.091)	0.067 (0.084)
Inc-bind Wald p	0.0046	0.0528	0.7751	0.7825
Inc-becomes (post)	-0.014 (0.004)	-0.226 (0.034)	-0.200 (0.040)	-0.217 (0.038)
Inc-becomes (pre)	-0.003 (0.007)	0.045 (0.032)	0.059 (0.054)	0.058 (0.051)
Inc-becomes Wald p	0.2930	0.0007	0.0003	0.0004
Inc-nobind (post)	-0.004 (0.006)	0.068 (0.070)	0.050 (0.056)	0.047 (0.054)
Inc-nobind (pre)	0.053 (0.025)	0.207 (0.133)	-0.423 (0.292)	-0.364 (0.267)
Inc-nobind Wald p	< 0.0001	< 0.0001	0.0050	0.0009
<i>Diagnostics</i>				
N	188,846	172,497	188,838	188,846
R^2	0.825	0.963	0.959	0.961
Within R^2	0.001	0.002	0.001	0.001

Notes: Cluster-robust standard errors in parentheses. Post = mean of β_k for $k = 0, \dots, 5$; pre = mean for $k \leq -2$. Wald p is the cluster-robust joint test of $H_0: \beta_k = 0$ across pre-event k . Per-cell decomposition of the muni-level realized-dose ES; each cell is run as a separate regression on {NC, cell}. Inc-bind is the load-bearing cell, with clean pre-trends on the three count outcomes. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

C Robustness for the Pooled Specification

The battery of Section 6, run on the pooled Increase specification of Equation (4). The school-district co-treatment, controls, fixed effects and municipality clustering are unchanged throughout, and the reported coefficient is β_{Inc}^M in every table.

C.1 Pre/Post Act 32

Table 18: Pooled Increase effect, split at the Act 32 enforcement break

	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Pre-Act 32 cohorts (59 munis)	-0.047** (0.021)	-0.341*** (0.111)	-0.237** (0.099)	-0.301*** (0.109)
Post-Act 32 cohorts (17 munis)	-0.001 (0.009)	-0.244*** (0.049)	-0.207*** (0.053)	-0.208*** (0.050)

Notes: Cohorts split at Act 32's effective year, 2012. Specification otherwise unchanged. SE in parens.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Act 47 Distressed-Municipality Exclusion

Table 19: Pooled Increase effect, excluding Act 47 distressed municipalities

	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
All Inc munis	-0.021* (0.011)	-0.284*** (0.052)	-0.220*** (0.052)	-0.248*** (0.052)
Drop 5 Act 47 munis	-0.037** (0.014)	-0.327*** (0.081)	-0.184** (0.075)	-0.233*** (0.079)

Notes: Act 47 status from DCED's distressed register. Excluded from the treated group only. SE in parens. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Pre-Trend Battery

Table 20: Pre-trend battery for the pooled Increase specification

	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Joint Wald p (leads)	0.259	0.029	0.060	0.103
Leads significant at 5% (of 4)	0	1	1	0
Smallest lead p	0.147	0.017	0.047	0.051
Linear pre-slope	0.013	-0.016	-0.084	-0.067
Linear pre-slope p	0.092	0.734	0.243	0.333
Clean	yes	no	no	yes

Notes: Tests on the four pre-event leads of the pooled realized-dose event study. Clean means the joint Wald does not reject at 10% and no single lead is significant at 5%.

C.4 Fake-Timing Placebo

Table 21: Pooled Increase effect at placebo event dates

	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
<i>Placebo event at $c - 3$ (37 eligible munis)</i>				
Pooled Inc	0.004	-0.052	-0.113**	-0.109**
	(0.006)	(0.040)	(0.054)	(0.049)
<i>Placebo event at $c - 4$ (31 eligible munis)</i>				
Pooled Inc	0.009	-0.052	-0.139**	-0.128**
	(0.007)	(0.057)	(0.064)	(0.060)
<i>Placebo event at $c - 5$ (27 eligible munis)</i>				
Pooled Inc	0.002	-0.099	-0.137*	-0.135*
	(0.010)	(0.074)	(0.081)	(0.076)

Notes: Placebo events 3, 4 or 5 years early, using only data before the real change, so every coefficient should be null. Counts fall with the shift because eligibility needs two placebo pre-years inside the panel. Panels overlap. SE in parens. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Robustness: Full Tables

Full tables behind the robustness summaries in Section 6.

D.1 Pre/post Act 32 split

Headline cell $\times$ dose specification re-estimated on pre-Act-32 ($c < 2012$) and post-Act-32 ($c \geq 2012$) cohort subsamples at the municipality level.

Table 22: Pre-/post-Act-32 split — municipality level

Outcome	Share live in muni	Log(work in own muni)	Log(in-commuters)	Log(total jobs)
Inc-bind				
Pre-Act 32	-0.068** (0.029)	-0.544*** (0.201)	-0.337** (0.166)	-0.432** (0.173)
Post-Act 32	-0.043*** (0.005)	-0.531*** (0.162)	-0.362** (0.150)	-0.415*** (0.147)
Inc-becomes				
Pre-Act 32	-0.043* (0.023)	-0.296*** (0.107)	-0.218** (0.109)	-0.276** (0.119)
Post-Act 32	0.004 (0.010)	-0.222*** (0.047)	-0.209*** (0.056)	-0.203*** (0.052)
Inc-nobind				
Post-Act 32	-0.028 (0.027)	-0.110 (0.152)	0.145 (0.224)	0.114 (0.196)
<i>Pre-Act 32</i>				
<i>N</i>	187,663	171,302	187,655	187,663
<i>R</i> ²	0.826	0.963	0.959	0.961
Within <i>R</i> ²	0.001	0.002	0.001	0.001
<i>Post-Act 32</i>				
<i>N</i>	187,479	171,136	187,471	187,479
<i>R</i> ²	0.826	0.963	0.959	0.961
Within <i>R</i> ²	0.001	0.002	0.001	0.001

Notes: Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$

(Pre-Act 32) vs $c \geq 2012$ (Post-Act 32). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.